MATH1090/1098
(Honours) Introduction to Set Theory

The Chinese University of Hong Kong

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Chapter 0

Advice to the student

- Sections marked $\star$ are optional for MATH1090 students.
- Bring a paper and pencil / pens to class, and leave your phone / laptop / tablet in your bag.
- During the in-class exercises, work with the people next to you. Ask lots of questions! You will learn better if you say your thoughts out loud.
- This class will have graded assignments and optional exercises. However, most of your grade comes from the midterm and final marks. The assignments are mainly there to help you practice. We suggest you spend 10 minutes every day doing a few optional exercises; this is the best way to internalize the subject.

Chapter 1

Logic

1.1 The general shape of a mathematical argument

This first chapter is about the different kind of arguments used in mathematics. Very roughly, these look like

$$\begin{array}{c} \text{Hypothesis A} \\ \text{Hypothesis B} \\ \hline \text{Conclusion C} \end{array}$$

or

$$\begin{array}{c} A \\ B \\ \hline \therefore C \end{array}$$

or in plain English: 'If A and B are true, then C is also true'. More concisely: 'C follows from A and B'.

We want to know:

Question 1. *What kind of statements can we use as hypotheses or conclusions?*

Question 2. *How do we deduce the conclusion from the hypotheses?*

Of course, in life there is no single correct answer to these questions. In mathematics, however, there is a more or less universal answer, which you will learn in the next few sections.

Propositions The simplest type of statement we will consider is a <u>proposition</u>, often denoted by a letter like A or B.

A proposition is a sentence or claim which can be either true or false. For example,

A: The moon is made of cheese.

and

B: I want new socks.

are propositions.[1] The sentence

Oh NOOOOO not again!!!

is not a proposition, because it is not a true/false claim.

1.2 Propositional logic.

Suppose you are investigating a murder. A, B, C, D, E, F, G, H represent your hypotheses, each of which can be either true or false; they are thus examples of propositions.

- A: the murder happened during the night.
- B: the murder happened during the day.
- C: Omar was asleep during the night.
- D: Michael was asleep during the day.
- E: Mykola was asleep between 3am and 11am.
- F: Omar is the murderer.
- G: Michael is the murderer.
- H: Mykola is the murderer.

These hypotheses are not independent. For instance:

1. If A is false, then B is true.
2. If B and D are true, then G is false.

We would like a good way to think about these dependencies, and to build complex statements out of them. Our tool will be propositional logic, also called zero-th order or Boolean logic.

[1] A good test: check if you can add 'It is true that...' or 'It is not true that...' to the beginning of the sentence.

Propositions as variables Let us be a little more formal. In propositional logic, we treat each proposition like a variable which can be either true (T) or false (F). The truth value of A is T if A is true, and F if A is false.

We thus forget all the irrelevant details of a proposition like A or D: Who was murdered? Who is Michael? Why was he sleeping during the day?. All we care about is whether A or D are TRUE or FALSE.

We write $A = T$ to indicate 'A is true' and $A = F$ to indicate 'A is false'. We call this the truth value, or simply the value, of the proposition.

Remark 1. *We will think of propositions as analogous to variables like a, b, c in algebraic equations like $a + b = b \cdot c$. Instead of taking numerical values like 5 or 1.23, propositions take only the values T or F, and are thus "simpler". We will discuss the analogues of $+$, $\cdot$, and $=$ below.*

1.3 The alphabet of propositional logic

Propositional logic is a kind of language, a bit like English or Chinese. It uses a very small alphabet of symbols. Here they are, labeled by their English names.

- $\wedge$ (AND)
- $\vee$ (OR)
- $\neg$ (NOT)
- $\leftrightarrow$ (IF AND ONLY IF or IS EQUIVALENT TO)
- $\rightarrow$ (IMPLIES)

A Boolean formula is a formula constructed using our propositions $A, B, C, ..., H$ together with the above symbols. For example, we can write

If A is false, then B is true

as the Boolean formula

$$\neg A \rightarrow B. \tag{1.1}$$

We can write

If B and D are true, then G is false

as

$$(B \wedge D) \rightarrow \neg G. \tag{1.2}$$

Exercise 1. *Can you write down some more logical dependencies among A, B, C, D, E, F, G, H as Boolean formulas?*

In our analogy, the Boolean operations like $\neg, \vee$ are like the symbols $+$ and $\cdot$ which we can use to take variables and form more complex algebraic formulas out of.

Boolean grammar We do not allow arbitrary strings of Boolean symbols when making Boolean formulas. For example,

$$A \vee \vee \vee \neg$$

is not a Boolean formula: it is just nonsense. There is a list of rules for making well-formed Boolean formulas, a bit like the grammar rules of a language. However, we will not spell out explicitly these rules explicitly. Instead, we trust that you will learn to do this intuitively through practice. If you find this confusing, ask us for some more examples.

Order of operations. Consider the formula

$$A \wedge B \vee C.$$

This formula is ambiguous! Does it mean 'Either A and B are both true, or C is true'? Or does does it mean 'A is true and either B or C' is true'?

To make it unambiguous, we add parenthesis. We write

$$(A \wedge B) \vee C$$

when we mean 'Either A and B are both true, or C is true', and we write

$$A \wedge (B \vee C)$$

when we mean 'A is true and either B or C' is true'. Now consider the formula

$$\neg A \wedge B.$$

Does it mean 'A is false and B is true' or 'It is not true that both A and B are true'? Here there is no ambiguity: by convention, we first apply $\neg$, then $\wedge$, so that the first reading is correct.

More generally, the conventional order in which to read a Boolean formula is: first $\neg$, then $\wedge$ and $\vee$, and finally $\rightarrow$ and $\leftrightarrow$. Any remaining ambiguities must be resolved by parentheses.

Different uses of the word <u>'proposition'</u> In most of mathematics, the word 'proposition' is usually reserved for a mathematical true/false claim that the author believes to be true (and has a proof of). This is different from its use within propositional logic. In practice, one can usually easily tell from context which meaning is being used.

1.4 Truth tables

Recall that each proposition $A, B, C, ..$ can have one of two values: true or false.

Boolean formulas like

$$\neg A, \quad A \wedge B \quad A \vee B$$

can also be either true or false, depending on whether A and B are true or false.

We can describe the conditions under which they are true or false using 'truth tables'. The left-hand columns below indicate possible values of A, B, whereas the right-hand columns indicate the resulting value of the formula.

A	$\neg A$
F	T
T	F

A	B	$A \wedge B$	$A \vee B$
F	F	F	F
F	T	F	T
T	F	F	T
T	T	T	T

A	B	$A \to B$	$A \leftrightarrow B$
F	F	T	T
F	T	T	F
T	F	F	F
T	T	T	T

Every Boolean formula thus determines a truth function whose input is a collection of values (either T or F) for each proposition appearing in the formula, and whose output is either T or F. In fact, if we wish to be totally precise, we may consider the truth tables above as definitions of the Boolean operations.

Equivalent formulas Just as in English and Chinese, we may formulate the same sentence in multiple ways, there can be many different Boolean formulas which express the same statement. More precisely, we will say two Boolean formulas A and B are logically equivalent if they determine the same truth function. In this case, we will write

$$A \equiv B$$

In the analogy with algebraic equations, logical equivalence should be considered the analogue of the $=$ sign.

The most immediate example of a logical equivalence is something common in natural language too: the double negative. If I say I am "not not cold" it means I am cold. Likewise, for any proposition A, we have

$$\neg(\neg A) \equiv A.$$

As a slightly less trivial example, we may observe that we can write $A \to B$ by referring only to "simpler" symbols $\neg$ and $\vee$:

$$A \to B \equiv (\neg A) \vee B.$$

That is, A implying B is the same thing as saying that either A is false or B is true (or both).

The following equivalences are common and important enough that we record them here:

1. Symmetry:
   - $A \wedge B \equiv B \wedge A$
   - $A \vee B \equiv B \vee A$
2. Associativity:
   - $A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$
   - $A \vee (B \vee C) \equiv (A \vee B) \vee C$
3. Distributivity:
   - $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$
   - $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$
4. de Morgan's laws:
   - $\neg(A \wedge B) \equiv (\neg A) \vee (\neg B)$
   - $\neg(A \vee B) \equiv (\neg A) \wedge (\neg B)$
5. Contrapositive: $A \to B \equiv (\neg B) \to (\neg A)$

Exercise 2. *Check that the above Boolean formulas are logically equivalent by inspecting their truth tables.*

Warning 1. *In general we need to be careful to distinguish between two forms of equivalence:*

$$A \leftrightarrow B \tag{1.3}$$

and

$$\phi \equiv \psi \tag{1.4}$$

The expression (1.3) *is a Boolean formula stating that the propositions A and B are equivalent. The expression* (1.4)*, on the other hand, is not a Boolean formula (the symbol $\equiv$ is not part of the Boolean alphabet!). It represents the statement that the Boolean formulae ϕ and ψ are logically equivalent.*

1.5 Propositional logic: rules of inference

Deduction (also called inference) is the art of constructing new knowledge from existing knowledge.

Consider the Boolean formulas

$$A \to B \tag{1.5}$$

and

$$B \to C. \tag{1.6}$$

If we believe that both of these formulas are true, then we can <u>deduce</u> that

$$A \to C \tag{1.7}$$

is true.

Here is another example. Consider

$$\neg A \tag{1.8}$$

and

$$B \to A. \tag{1.9}$$

If we believe that both formulas are true, then we can deduce that B is false. We write these deductions as

$$\begin{array}{c} A \to B. \\ B \to C. \\ \hline \therefore A \to C. \end{array}$$

and

$$\begin{array}{c} \neg A. \\ B \to A. \\ \hline \therefore \neg B. \end{array}$$

So far, we are speaking intuitively. More formally, what we mean is that <u>every assignment of truth values to A, B, C which makes the first two formulas true will also make the third formula true.</u> We can think of this like a compound version of the truth tables of the above section.

Exercise 3. *Check the above examples using truth tables. In other words, check that if B is true, then at least one of the two formulas* (1.8) *or* (1.9) *must be false.*

This is an example of logical deduction. Although this kind of deduction is denoted by the special symbol $\vdash$ or $\models$ in formal logic, in practice we usually simply write (in English) something like

It follows from (1.8) and (1.9) that B is false.

Here is another example. If we believe

$$(A \vee B) \wedge \neg B \tag{1.10}$$

is true, then we can deduce that A is true.

Exercise 4. *Check this!*

Exercise 5. *Show that*

$$A \to B \tag{1.11}$$
$$B \to C \tag{1.12}$$
$$C \to A \tag{1.13}$$

is equivalent to

$$A \leftrightarrow B \leftrightarrow C. \tag{1.14}$$

There are a finite number of rules that allow us to deduce new Boolean formulas from old ones. We will not learn all of these rules, but you can find them listed on e.g. Wikipedia. You should not try to memorize them, and for practical purposes, your common sense will be enough.

1.6 Tautology and contradiction

There are two kinds of 'extreme' Boolean formulas: those which are always true, and those which are always false.

Example 1. *The formula*

$$A \leftrightarrow A$$

is always true, regardless of the truth-value of A. The same holds for

$$A \to A$$

and

$$A \vee \neg A.$$

We call such a formula a <u>tautology</u>.

Example 2. *The formula*

$$A \wedge \neg A$$

is always false, regardless of the truth-value of A.

We call such a formula a <u>contradiction</u>.

1.7 Why are we learning this?

In practice, you will not see many Boolean symbols used in your math classes, nor will you find many Boolean symbols in math papers. Mathematical reasoning is usually communicated using a language like English or Chinese (supplemented with appropriate notation), not using the Boolean language. Boolean is the language of computers: writing things in Boolean typically makes them harder for us humans to understand! However:

Any mathematician should be able to translate their reasoning from their native language into Boolean.

Just because you can, however, does not mean you should – just as paying for a new car using pennies would be incredibly impractical and irritate other people. Once you have some practice with Boolean logic, you can go back to writing your arguments in English or Chinese (or Irish, Ukrainian, or Urdu). You may find that you can express yourself more clearly and less ambiguously, having internalized the basic rules of Boolean logic.

In fact, Boolean logic is not powerful enough to capture all mathematical arguments. In the next section, we will learn about first-order logic, which is powerful enough for almost all of modern mathematics. Contrary to Boolean logic, the symbols of first-order logic will be frequently used in your classes and in math papers.

1.8 Practical examples of propositional logic

See the worksheet.

1.9 Predicate logic.

Consider the following argument:

All humans are mortal.
Confucius is human.

Therefore, Confucius is mortal.

This argument is difficult or impossible to state using only propositional logic.

Instead, we need the more powerful tools of predicate logic, which we introduce in this section. We will introduce the ingredients of predicate logic one by one:

- Variables $x, y, z, \ldots$.
- Predicates $P, Q, R, \ldots$.
- Quantifiers $\forall, \exists$.

Variables Predicate logic deals with statements like 'All humans like S', or 'some humans dislike S'. In other words, we have a collection of things (humans in this example), and we are interested in whether some or all of them have some property. More generally, we want to make statements about things in some kind of collection of objects, without yet specifying which object we are talking about. We will use lower-case variables $x, y, z, ..$ to indicate the unspecified members of this collection.

Let's write $M(x)$ for 'x is mortal' and $H(x)$ for 'x is human'. Then we can write our original argument as

$$\begin{array}{l} \text{For all } x,\ H(x) \to M(x). \\ H(s). \\ \hline \therefore M(s). \end{array}$$

Predicates The word 'predicate' means 'property', and an important feature of predicate logic is that we can form a TRUE/FALSE proposition of the form

The object s has the property P.

For instance, let's write s for Confucius, and P for the property of being bearded. We write

$$P(s)$$

for the statement 'Confucius is bearded'. We can also apply M to other people: if we write p for Mencius, we can write

$$P(p)$$

for 'Mencius is bearded'.

Put another way: in propositional logic, the basic variables we considered were only capable of being either True (T) or False (F). In predicate logic, we can now consider another answer: it depends (on what x or some other variable is). We can think of predicates a little like "propositions depending on an arbitrary variable".

Quantifiers Why did we write 'for all x' in the first line? Why not just $H(x) \to M(x)$?

The formula $H(x) \to M(x)$ is neither true nor false: that depends on what x is. By adding 'for all x', we make it into a proposition which can be either true or false.

We use two special symbols for this, called quantifiers:

- $\forall$ (FOR ALL).
- $\exists$ (THERE EXISTS).

The formula

$$\forall x P(x)$$

means 'for all x, x has property P', whereas

$$\exists x Q(x)$$

means 'there exists x, such that x has property Q'. We can use these symbols to rewrite our original argument as

$$\begin{array}{l} \forall x(H(x) \to M(x)). \\ H(s). \\ \hline \therefore M(s). \end{array}$$

As with propositional logic, whenever we construct a sensible string of symbols using variables together with $\vee$, $\forall$, etc., we refer to it as a <u>predicate formula</u>. As before, we leave the grammar of this language to common sense.

Exercise 6. *Write the following as predicate formulas:*

1. *All couples have fights.*
2. *Everyone is reading at least two books.*
3. *My classmate has exactly one friend.*

The statement of "exactly one" existing is common enough in mathematics that we have a shorthand for it: $\exists!$. Using this symbol, we can also write the last statement as $\exists! x F(o, x)$.

Equivalence Just as we had a notion of logical equivalence in propositional logic, we also have a notion of logical equivalence between predicate formulas. However, it is slightly more complicated to define formally, so we will stick to our intuition when dealing with it: roughly speaking, two predicate formulas are logically equivalent if they hold for all possible substitutions.

Exercise 7. *Suppose P is a predicate. Rewrite*

$$\neg \forall x P(x)$$

using $\exists$.

Remark 2. *As a corollary, it is clear that $\exists x$ is the same thing as saying $\neg \forall x \neg$, and thus the symbols $\exists$ and $\forall$ are not independent.*

Consider the following argument:

My friend does not like cheese.
My friend is human.

Therefore, not all humans like cheese.

We write $C(x)$ for 'x likes cheese'. Then this becomes

$$\frac{\exists x(H(x) \wedge \neg C(x)).}{\therefore \neg \forall x(H(x) \to C(x)).}$$

Exercise 8. *Find a few examples of laws of predicate logic.*

Of course, we can perfectly well allow our predicates to depend on multiple variables as well. For example, we could have the predicate $D(x, y)$ meaning "x dislikes y", or $L(x, y, z)$ meaning "x likes y with z on top".

Here are some important examples of equivalences in predicate logic:

1. Distributivity:
   - $\forall x(P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$
   - $\exists x(P(x) \vee Q(x)) \equiv (\exists x P(x)) \vee (\exists x Q(x))$
2. Quantifier negation laws:
   - $\neg \forall x P(x) \equiv \exists x \neg P(x)$
   - $\neg \exists x Q(x) \equiv \forall x \neg Q(x)$
3. de Morgan's laws:
   - $\exists x \neg(P(x) \vee Q(x)) \equiv \exists x(\neg P(x) \wedge \neg Q(x))$
   - $\forall x \neg(P(x) \wedge Q(x)) \equiv \forall x(\neg P(x) \vee \neg Q(x))$
4. Permutation:
   - $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
   - $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$

Exercise 9. *Translate the following sentences into English, and explain the difference between them.*

$$\forall x \exists y(x < y)$$

and

$$\exists y \forall x(x < y).$$

In particular, we see that permutation does ***not*** *hold for mixed quantifiers:*

$$\forall x \exists y P(x, y) \text{ and } \exists y \forall x P(x, y)$$

are not logicaly equivalent.

Exercise 10. *List some variants of the equivalences listed above which fail.*

Remark 3. *Note that predicate logic is not a separate system from propositional logic, but rather an enhancement of it. To recover propositional logic, roughly speaking we think of propositions as those predicates which have no dependence on variables, and ban all usage of the symbols* $\forall$ *and* $\exists$*.*

1.10 Practical examples of predicate logic

See the worksheet.

Chapter 2

Sets

Armed with predicate logic, we need one more ingredient to complete the language which we need to do almost all of mathematics. A good reference for this material is the book Naive Set Theory by Paul R. Halmos.

2.1 Naive set theory

Sets A set is a collection of things.

Example 3. *$A = \{$Peng Chau, Cheung Chau, Lantau$\}$, $B = \{1, 2, 3\}$ and $C = \{$Chinese, English$\}$ are sets.*

Mathematics is written in the language of sets. We write

$$x \in A$$

if x is contained in A. We call x an element of B. Thus, continuing the previous example, $1 \in B$. We write

$$x \notin B$$

if x is not in B. Thus $5 \notin B$.

We usually write elements in lowercase, and sets in uppercase. But we sometimes deviate from that convention.

Warning 2. *A multiset is a collection of things (just like a set), except that a multiset can contain multiple copies of a single thing, and it remembers how many copies it contains. For example, $A = \{1, 2, 3\}$ and $B = \{1, 1, 2, 3\}$ represent two different multisets. In this class, we will only be interested in sets, and almost never work with multisets.*

We write $A = B$ if the sets A and B have the same elements. We say A and B are equal. As far as set theory goes, there is no difference between A and B. We record this by the following grandiose-sounding 'axiom':

Axiom 1 (The Axiom of Extension)**.** *$A = B$ if and only if*

$$\forall x(x \in A \leftrightarrow x \in B).$$

Remark 4. *Wait a second - what exactly is a set? At this stage, you may complain: 'a collection of things' doesn't sound very precise! What kind of things? What kind of collections? What is a thing or a collection anyway? How can we do mathematics starting from such a vague concept?*

There are two answers to this question. The first is that, with a bit of practice, you will develop a very precise intuition for what is a set and what isn't.

If the first answer does not satisfy you, then congratulations! You are a formalist. To work formally with sets, we will write down nine 'axioms', such as the axiom of extension above. These axioms will tell us how to build and characterize the sets we need using expressions like $x \in X, y \notin X$. We need only these axioms, together with the language of predicate logic, to do all the set theory we will ever need.

Since this class is not an introduction to axiomatic logic, we will not cover all of the nine axioms. But we will see a few of them as we go.

Inclusions of sets: We write

$$A \subset B$$

if every element of A is contained in B, and call A a subset of B. We call A a strict subset of B if in addition, B contains an element not in A.

Exercise 11. *Prove that $A \subset B$ and $B \subset A$ implies $A = B$.*

Warning 3. *Some authors write $A \subset B$ instead of $A \subset B$, and write $A \subset B$ only when A is a strict subset.*

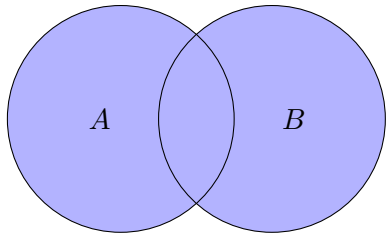
2.2 Making new sets

Even if we have our hands on some nice examples of sets, as greedy people we want to have more. To resolve this, we may construct new sets from old ones using various operations. We have:

1. $A \cup B$ (UNION):
   The union of two sets A and B is the set containing all elements that are in A or in B (or in both):

   $$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

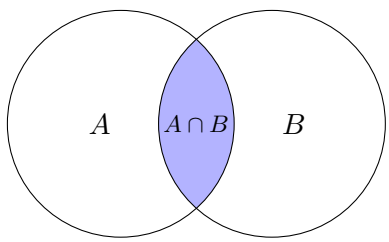
   Venn diagrams are often used to visually represent relationships between sets. For instance, the blue region below visualises the union of two sets.



2. $A \cap B$ (INTERSECTION):
   The intersection of two sets A and B is the set containing all elements that are in both A and B:

   $$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

   The Venn diagram visualizing the intersection looks like this.



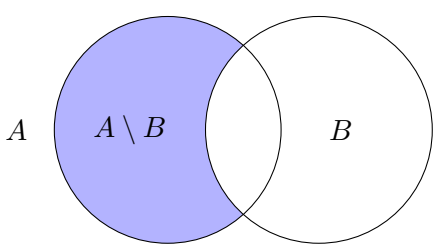
   Two sets are disjoint if $A \cap B = \emptyset$. What would the corresponding Venn diagram look like in this case?

3. $A \setminus B$ (COMPLEMENT and SET DIFFERENCE): The complement of a set $B \subset A$ is the set containing all elements of A that are not in B:

   $$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

   More generally, the set difference of two sets A, B, (not necessarily subsets)

   The Venn diagram visualizing the set difference looks like this.



4. $A \times B$ (PRODUCT):
   The Cartesian product of two sets A and B, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$:

   $$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

   For example, $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ is the Cartesian plane.

   Exercise 12. *Let the set A have exactly 5 elements, and the set B exactly 3 elements. How many elements does $A \times B$ have?*

   More generally, the product of n sets $A_1, A_2, \ldots, A_n$ is:

   $$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for each } i\}.$$

5. A^n (n-FOLD PRODUCT):
   As a special case of the above, the n-fold Cartesian product of a set A with itself n times is denoted A^n as a shorthand:

   $$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ times}} = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A \text{ for all } 1 \leq i \leq n\}.$$

   For instance, $\{0, 1\}^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}$.

6. $P(A) = 2^A$. (POWER SET):
   The power set of a set A, denoted $\mathcal{P}(A)$ or 2^A, is the set of all subsets of A:

   $$\mathcal{P}(A) = \{X \mid X \subset A\}.$$

   If A has n elements, then $\mathcal{P}(A)$ has 2^n elements. For example,

   $$\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$$

   The notation 2^A arises because there is a natural bijection between the power set and the set of all functions from A to the two-element set $\{0, 1\}$.

Disjoint union* $A \sqcup B$ (DISJOINT UNION):
The disjoint union of two sets A and B is given by $A \sqcup B = (A \times \{1\}) \cup (B \times \{2\})$. More generally, we can write the disjoint union of n sets A_n:

$$\bigsqcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} \{(x, i) \mid x \in A_i\}$$

The additional index is effectively "tagging" each element with its set of origin, so that we can remember which set it originally came from.

The Axiom of Comprehension Recall that in our discussion of predicate logic, we were somewhat vague about where variables could actually take values. For the purposes of this course, we will take them to represent elements of sets. With this in mind, we can state:

Axiom 2 (Axiom of Comprehension)**.** *Let P be a predicate and S a set. Then*

$$\{x \in S \mid P(x)\}$$

is a set.

2.3 The algebra of sets

Lemma 1. *Let A, B, C be sets. Then*

- $A \cup \emptyset = A$
- $A \cup B = B \cup A$ *(commutativity)*
- $A \cup (B \cup C) = (A \cup B) \cup C$ *(associativity)*
- $A \cup A = A$ *(idempotence)*
- $A \subset B$ *if and only if* $A \cup B = B$

Exercise 13. *State and prove the analogous properties for intersections.*

Lemma 2. *(Distributive laws) Let A, B, C be sets. Then*

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Exercise 14. *Prove or disprove the following statement:*
The equality $(A \cap B) \cup C = A \cap (B \cup C)$ holds true if and only if C is a subset of A.

Often, one is in a situation where everything is happening in some particular set (for example, our daily lives take place in $\mathbb{R}^3$). Let us therefore assume that all the sets below are subsets of a universal set E (here E stands for "everything"). Then for a set A, we denote by A^c the complement of A in E, i.e. $A^c = E \setminus A$ (note that the set E is dropped from the notation, so you should only use this when it is obvious).

Lemma 3. *Suppose $A \subset E$. Then:*

- $(A^c)^c = A$

- $\emptyset^c = E$, $E^c = \emptyset$
- $A \cap A^c = \emptyset$, $A \cup A^c = E$
- $A \subset B$ *if and only if* $B^c \subset A^c$

Lemma 4. *Suppose $A \subset E$ (de Morgan's laws)*

- $(A \cup B)^c = A^c \cap B^c$
- $(A \cap B)^c = A^c \cup B^c$

2.4 Functions

We will start with an informal discussion of functions, and then make it more precise using sets.

Informally, a function[1] f is a rule that takes an input[2] x and returns an output[3] $f(x)$.

Example 4. *You have learned many examples of functions in school, such as*

$$f(x) = x^2$$

and

$$g(x) = \sin(x).$$

The domain of f is the set of possible values of x. The target of f is the set of possible values of f. We write

$$f : X \to Y.$$

for a function with domain X and target Y.

The most important feature of a function is that it returns a unique output for a given input.

Warning 4. *Some people call such things single-valued functions, and also allow things called multivalued functions, which have multiple outputs. An example of such a multivalued function might be $\sqrt{x}$, with*

$$\sqrt{4} = \pm 2.$$

We will reserve the word function for single valued functions, and we will not use the term 'multivalued function' in this class.

[1] Synonyms: map, mapping
[2] Synonym: argument
[3] Synonym: value

The graph of f is the subset of $X \times Y$ defined by

$$\{(x, y) \mid y = f(x).\}$$

Knowing the graph is the same as knowing the function.

Question 3. *What kind of subsets of $X \times Y$ are graphs of functions?*

Here is one property satisfied by all graphs. Let f be a function. Then for each $x \in X$, there is a unique $y \in Y$ such that (x, y) lies in the graph of f. In fact, any subset $R \subset X \times Y$ satisfying this condition is the graph of a function: we can simply define $f(x)$ to be the unique y such that $(x, y) \in R$.

In the next subsection, we will turn this around and use graphs to give a rigorous definition of functions.

Set-theoretic definition of functions

Definition 1. *Let X, Y be sets. A function with domain X and target Y is a subset*

$$f \subset X \times Y$$

such that for all $x \in X$, there exists a unique $y \in Y$ such that $(x, y) \in f$. We write

$$y = f(x)$$

if $(x, y) \in f$.

We sometimes write $f : X \to Y$ and/or use the notation

$$x \to f(x).$$

Exercise 15. *Given sets A and B, show that there exists a set B^A consisting of all functions with domain A and target B.*

Examples of functions Every set comes with a seemingly trivial but very important function

$$\mathrm{id}_X : X \to X$$

which sends $x \mapsto x$, called the identity on X.

Exercise 16. *Give some other examples of functions.*

Image and preimage: Consider a function $f : X \to Y$. There are several closely related usages of the word image:

- image of an element: For $x \in X$, $f(x)$ is the image of x.
- image of a subset: For $A \subset X$,
$$f(A) = \{f(x) \mid x \in A\} \subset Y.$$
- image of f: The set $f(X)$ of all actual outputs:
$$f(X) = \{y \in Y \mid \exists x \in X \text{ with } f(x) = y\}$$
- preimage:[4] For $B \subset Y$,
$$f^{-1}(B) = \{x \in X \mid f(x) \in B\} \subset X.$$
Note: $f^{-1}(B)$ exists even if f is not invertible.
- For a single element $y \in Y$, we write $f^{-1}(y) = f^{-1}(\{y\})$.

Composition: Given $f : X \to Y$ and $g : Y \to Z$, the **composition** $g \circ f : X \to Z$ is defined by:
$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$
Composition is associative: $h \circ (g \circ f) = (h \circ g) \circ f$ when defined.

Guessing the domain of a function In practice, a function is often given without specifying its domain explicitly. For instance, consider the function $\frac{1}{x}$. This function takes as input a number, and outputs a number. Depending on context, this might be a rational number or a real number. However, in all of these cases, it does not take the argument $x = 0$; we say it is not defined at 0.

So we might interpret $f = \frac{1}{x}$ as a function with domain $\mathbb{R} \setminus 0$ or $\mathbb{Q} \setminus 0$, depending on context.

Injectivity, surjectivity, bijectivity: $f : X \to Y$ is injective or one-to-one if distinct inputs yield distinct outputs:
$$\forall x_1, x_2 \in X,\ f(x_1) = f(x_2) \to x_1 = x_2.$$
Equivalently, $\forall y \in Y$, $f^{-1}(y)$ contains at most one element.

$f : X \to Y$ is surjective or onto if every element of Y is an output:
$$\forall y \in Y,\ \exists x \in X \text{ such that } f(x) = y.$$
Equivalently, $f(X) = Y$. Finally, $f : X \to Y$ is bijective if it is both injective and surjective.

Exercise 17. *Give some examples and non-examples of injective and surjective functions.*

[4] Synonyms: inverse image, fibre

Inverses: For $f : X \to Y$, bijective we can define the notion of an inverse function. We say an <u>inverse function</u> $g : Y \to X$ exists if and only if there is a function g, denoted f^{-1}, such that

- $f^{-1} \circ f = \mathrm{id}_X$ (left inverse).
- $f \circ f^{-1} = \mathrm{id}_Y$ (right inverse).

Exercise 18. *Show that the inverse function of f, if it exists, is unique.*

Exercise 19. *Show that an inverse function of f exists if and only if f is a bijection.*

Remark 5. *Beware of a slight notational overlap. The symbol f^{-1} has two meanings:*

- *For any function $f : X \to Y$, $f^{-1}(B)$ denotes the preimage of $B \subset Y$.*
- *For a bijective function f, f^{-1} also denotes the inverse function.*

The interpretation should essentially always be clear from the context.

2.5 Cardinality

<u>Cardinality</u> is a fancy way of saying 'size'.

Definition 2. *Two sets X and Y <u>have the same cardinality</u> if there is exists a bijection*

$$f : X \to Y.$$

We write

$$|X| = |Y|.$$

We will see later that this concept is more subtle than it appears.

2.6 Operations

Definition 3. *Let S be a set. A function $f : S^n \to S$ is called an n-ary operation.*

Depending on the context, there are different standard notations for such things:

Example 5. *Addition of numbers is a 2-ary operation, which we write as*

$$(a, b) \to a + b.$$

Example 6. *Exponentiation of numbers is a 1-ary operation, which we write as*

$$x \to e^x.$$

Example 7. *We can even talk about 0-ary operations: these are functions to S with no arguments at all! That is, a 0-ary operation*

$$f : \emptyset \to S$$

can be thought of as just an element of S.

2.7 Relations

A relation on a pair of sets X, Y is a subset

$$R \subset X \times Y.$$

We write

$$xRy$$

to mean

$$(x, y) \in R.$$

Example 8. *Let X be the set of countries and Y be the set of cities in the world. Then*

$$\{(x, y) \in X \times Y \mid y \text{ is the capital city of } x\}$$

is a relation. Whether it is a function depends on which era we are in, as many states had summer capitals historically; thanks to air conditioning, it is most likely a function today.

If $X = Y$, we simply refer to it as a relation on X (or Y) . An example of a relation on $\mathbb{Z}$ is

$$\left\{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid y^2 = x.\right\}$$

The domain of R is the set of $x \in X$ such that there exists $y \in Y$ with xRy. The image of R is the set of $y \in Y$ such that there exists $x \in X$ with xRy.

Exercise 20. *Find the domain and range of the above relation.*

Clearly, a function is a special case of a relation, but with the restriction that a given element in X can only be related to one element of Y.

Example 9. *Show that the relation on $\mathbb{R}$*

$$\left\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x^2 + y^2 = 1\right\}$$

does not define a function.

2.8 Relations from a set to itself

Relations $R \subset X \times X$ from a set to itself play a special role. Here are some extra properties they can satisfy.

Definition 4. *We say a relation $R \subset X \times X$ is*

- *transitive if*

$$\forall x, y, z : xRy \wedge yRz \to xRz.$$

- *reflexive* if
$$\forall x : xRx.$$
- *symmetric* if
$$\forall x, y : xRy \leftrightarrow yRx$$
- *antisymmetric* if
$$\forall x, y : xRy \wedge yRx \rightarrow x = y.$$

Exercise 21. *Come up with a few examples of relations which satisfy and fail each property.*

2.9 Partial orders

A very important example of a relation is the following.

Definition 5. *A relation $R \subset X \times X$ is a partial order*[5] *if it is reflexive, antisymmetric, and transitive.*

In this case, we usually write $x \preceq y$ instead of xRy, and $x \prec y$ if $x \preceq y$ and $x \neq y$.

Example 10. *Let $X = \{a, b, c\}$ and $P = \mathcal{P}(X)$. Define:*

$$A \preceq B \quad \textit{if and only if} \quad A \subset B$$

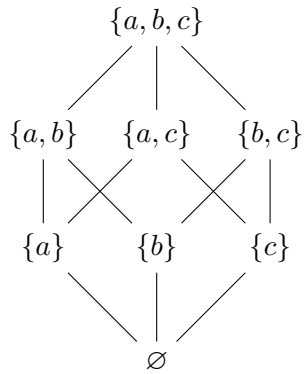
It is easy to see that the axioms are satisfied.

When feasible, we can depict a partial order using Hasse diagram. This is a graphical representation of a partial order where:

1. Elements are drawn as points (nodes)
2. If $a \prec b$ (i.e., $a \preceq b$ and $a \neq b$), then b appears **above** a
3. A line is drawn from a to b if $a \prec b$ and there is **no** c such that $a \prec c \prec b$ (i.e., b *covers* a)
4. All other relations are implied by transitivity (so we don't draw them)

We may draw the Hasse diagram of the above example as:

[5] Also partially ordered set, or poset for short.



Exercise 22. *Let $P = \{1,2,3\}$ and define $a \preceq b$ if $a \leq b$ (usual order)* ***except*** *also define $1 \preceq 3$ and $3 \preceq 1$. Show that this is not a partial order: which property does it fail to satisfy?*

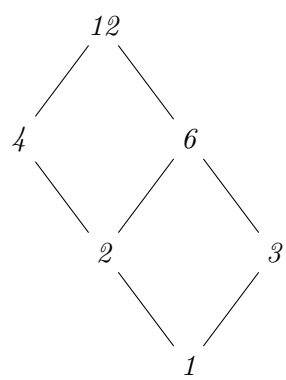
Example 11. *Let $X = \mathbb{Z}_+$ (positive integers) and define:*

$$a \preceq b \quad \textit{if and only if} \quad a \mid b \ (a \textit{ divides } b)$$

We check that this is a partial order:

- *<u>Reflexive:</u> $a \mid a$ since $a = a \cdot 1$*
- *<u>Antisymmetric:</u> If $a \mid b$ and $b \mid a$, then $a = b$ (for integers, up to sign, but here all positive)*
- *<u>Transitive:</u> If $a \mid b$ and $b \mid c$, then $a \mid c$ (try writing this out in predicate logic)*

Example 12. *Let $X = \{1,2,3,4,6,12\}$ and with the divisibility partial order as above. The Hasse diagram can be drawn as:*



2.10 Equivalence relations

The other fundamental example of a relation is that of an equivalence relation.

Definition 6. *A relation $R \subset X \times X$ is an* <u>*equivalence relation*</u> *if it is reflexive, symmetric, and transitive.*

Example 13. *Let $X = \{$people in this class$\}$ and define xRy if x and y live in the same building. Then R is an equivalence relation.*

Exercise 23. *Give an example of something that fails one of each property but satisfies the other two.*

Definition 7. *Let $m \in \mathbb{Z}$, $m > 0$. For integers $a, b \in \mathbb{Z}$, we say a and b are* <u>*congruent modulo m*</u>*:*

$$a \equiv b \pmod{m}$$

if $m \mid (a - b)$ (i.e., $a - b$ is divisible by m).

Equivalently: $a \equiv b \pmod{m} \iff a = b + km$ for some $k \in \mathbb{Z}$.

Exercise 24. *Show that congruence modulo m is an equivalence relation.*

Example 14.

$$a \equiv 0 \pmod{2} \leftrightarrow a \text{ is even}$$
$$a \equiv 1 \pmod{2} \leftrightarrow a \text{ is odd}$$

Definition 8. *Let $R \subset X \times X$. An* <u>*equivalence class*</u> *is a nonempty subset $S \subset X$ such that*

$$x \in S \wedge xRy \to y \in S$$

and

$$x, y \in S \to xRy.$$

In other words, an equivalence class is a subset of elements of X, all of whom are related to eachother, and with no "missing" related elements.

Definition 9. *Let R be an equivalence relation on X, and let $x \in X$. We define the* <u>*equivalence class of x*</u> *to be the subset $[x]_R \subset X$ given by*

$$\{y \mid xRy\}.$$

By reflexivity, it is immediate that for any $x \in X$, $x \in [x]_R$.

Proposition 1. *A set with an equivalence relation is the union of its equivalence classes, all of which are pairwise disjoint.*

Proof. Let $x \in X$. Then $x \in [x]_R$. This shows that X is a union of equivalence classes.

We now show that any two equivalence classes are equal or disjoint. Suppose $S, T \subset X$ are equivalence classes. Let $x \in S \cap T$. Let $y \in S$. Then xRy (because S is an equivalence class) and therefore $y \in T$ (because T is an equivalence class). This shows $S \subset T$. By the same argument, $T \subset S$ and thus $T = S$. $\square$

We sometimes say that the equivalence classes <u>partition</u> the set.

As a matter of notation, an equivalence relation R is often denoted instead by $\sim$, so that xRy is written $x \sim y$. The next definition will be very useful later.

Definition 10. *Let X be a set, and $\sim$ an equivalence relation on it. The set of equivalence classes is called the <u>quotient set</u>, denoted by $X/\sim$.*

Chapter 3

The integers

3.1 A mathematician's view of the integers

Consider the following sentence:

Any odd integer greater than 5 may be expressed as the sum of three primes.

We want to think about such statements using the language of logic and set theory. Our first task is to understand the set of all natural numbers. We make the following informal definition:

Definition 11. *Let*

$$\mathbb{N} = \{0, 1, 2, 3, 4, 5, ...\} \tag{3.1}$$

be the set of all natural numbers.

This definition is informal because we have not said exactly what the elements of this set are. Instead, we use the notation ... to suggest that aside from $0, 1, 2$, we should also include other elements (such as 3 and 4), and we think the reader knows how to guess these elements.

Another seemingly frivolous objection: different people may use different symbols for numbers:

Definition 12. *Let*

$$N = \{0, I, II, III, IV, V, ...\}$$

be the set of all Roman numbers.

Of course, this is not a practical problem : we can easily translate between the two. A more serious issue: how would you explain to a computer, or an alien visiting from the rings of Saturn, what those mysterious (...) are hiding?

In the end, the main problem with our informal definition is that it does not reveal what makes the natural numbers the natural numbers. It does not capture their essence!

3.2 A formal definition: Peano's axioms

Let's try again. Suppose $\mathcal{N}$ is a set together with:

- An element $0 \in \mathcal{N}$.
- A function $S : \mathcal{N} \to \mathcal{N}$.

Definition 13. *We say the triple of data $(\mathcal{N}, 0, S)$ (or just $\mathcal{N}$ for short) is a* <u>*model of the natural numbers*</u> *if:*

1. *If $S(x) = S(y)$ for $x, y \in \mathcal{N}$, then $x = y$ (S is injective).*
2. *For all $x \in \mathcal{N}$, $S(x) \neq x$.*
3. *There is no $x \in \mathcal{N}$ such that $S(x) = 0$.*
4. *(Principle of Induction) Suppose a predicate[1] P is true of 0, and that whenever P is true of x, it is also true of $S(x)$. Then P is true for all $x \in \mathcal{N}$.*

Exercise 25. *Show that the set $\{0, 1, 2\}$ with $S(0) = 1$, $S(1) = 2$, $S(2) = 0$* <u>*does not*</u> *satisfy all Peano axioms. Which axioms fail?*

We are used to thinking of these as properties or structures of the natural numbers. Now we will reverse our perspective and think of them as a <u>definition</u> of the natural numbers.

Compared with our informal definition, this has a number of drawbacks. It is hard to understand at first, and uses unfamiliar notation. It seems overly complicated for the apparently simple task of defining the natural numbers.

It has the benefit, however, of being completely defined in terms of sets and predicate logic. No magical intuition or fundamentally new concepts are required, and it becomes a construction reducible to more primitive ingredients.

3.3 Defining arithmetic operations: $+$ and $\cdot$

We would like to define the usual operations we are familiar with on integers. First, we define $+$:

$$\begin{cases} a + 0 = a \\ a + S(b) = S(a + b) \end{cases}$$

and likewise with the order reversed.

This is a recursive definition: we begin with a "base case" of adding 0 to itself or to other numbers, and to add something to a we first express it as a successor, then reduce the problem to computing the successor of a "smaller" number, $a + b$. Repeatedly applying this definition, we eventually get back down to 0.

[1] Strictly: a unary predicate, that is, with one argument.

Exercise 26. *Show this formally using Peano's axioms.*

It is instructive to compute an example to see exactly how this works. Let's try to catch up with elementary school students, and show $2 + 3 = 5$.

Example 15. *Let* $2 = S(S(0))$, $3 = S(S(S(0)))$. *Then:*

$$\begin{aligned} 2 + 3 &= S(S(0)) + S(S(S(0))) \\ &= S(S(S(0)) + S(S(0))) \\ &= S(S(S(S(0)) + S(0))) \\ &= S(S(S(S(S(0)) + 0))) \\ &= S(S(S(S(S(0))))), \end{aligned}$$

a number commonly denoted as 5.

Exercise 27. *Show that for all* $n \in \mathbb{N}$, $0 + n = n$.

As with addition, we define $\cdot$ recursively too:

$$\begin{cases} a \cdot 0 = 0 \\ a \cdot S(b) = a + (a \cdot b) \end{cases}$$

This is well-defined again by the principle of induction.

Exercise 28. *Calculate* $2 \cdot 3$.

Exercise 29. *Show from the definitions that addition and multiplication are commutative and associative.*

Exercise 30. *Show that for any* $a, b, c \in \mathbb{N}$, *we have the distributivity property:*

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

3.4 Principle of Induction

Having played around with some of the consequences, it is hopefully now clear what we have done: we have taken the idea of mathematical induction and turned it into a definition. In essence, this is the only property that one ever needs or uses about the natural numbers. It is less important to us what they actually are, and more important how they are related to one another, in order that we may prove properties. This is more generally a common principle of (much of) modern mathematics, and you should contemplate it.

Exercise 31. *Let $M = \{0, 1, 2, \ldots\} \cup \{a, b, c\}$, with the usual successor function*

$$S(0) = 1, S(1) = 2, \ldots$$

for the numerical elements, and define

$$S(a) = b, \quad S(b) = c \quad S(c) = a.$$

Show that $(M, 0, S)$ satisfies all of the conditions to be a model of the natural numbers <u>*except*</u> *the principle of induction.*

3.5 From $\mathbb{N}$ to $\mathbb{Z}$

At some point, one encounters equations in which only natural numbers appear, like $1 = x + 2$, but which have no solution in the natural numbers. We would like to extend our number system to allow for these more exotic "numbers" which, as the naming suggests, are not quite as natural (I have certainly never seen -1 apples)[2].

To do this, we will first take a set that is too big, and put an equivalence relation on it. The set of equivalence classes will then be the thing we are looking for. That sounds rather elaborate, but the motivation for doing this is the following: we are trying to represent "the solution to $1 = x + 2$". However, whatever this thing is, it is also the solution to $2 = x + 3$, and the solution to $3 = x + 4$. We can therefore try to think of this entire collection of equations, indexed by $(1, 2), (2, 3), (3, 4), \ldots$ as representing the same thing, and <u>define</u> that to be -1.

Definition as a quotient set More precisely, and formally, we consider an equivalence relation $\sim_{\mathbb{Z}}$ on the set $X = \mathbb{N}^2$ given by

$$(a, b) \sim_{\mathbb{Z}} (c, d) \text{ if } a + d = b + c.$$

Exercise 32. *Check that this is an equivalence relation.*

We define $\mathbb{Z} := X/\sim_{\mathbb{Z}}$, the set of equivalence classes:

$$[(a, b)] := \{(c, d) \in X \mid (c, d) \sim_{\mathbb{Z}} (a, b)\}.$$

Thus, an <u>integer</u> is an equivalence class of the relation $\sim_{\mathbb{Z}}$ on $X = \mathbb{N}^2$.

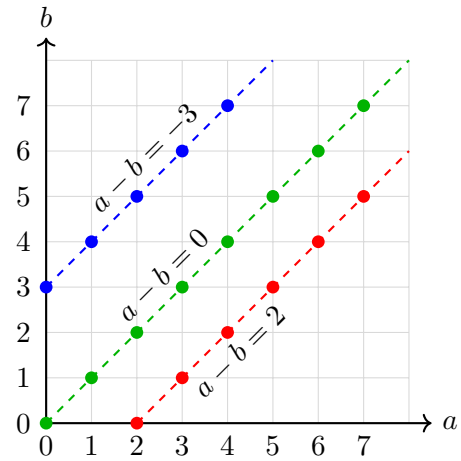
This looks a little bit funny. Aren't the integers supposed to contain the naturals? Where are they?

In fact, we can embed the familiar $\mathbb{N}$ into $\mathbb{Z}$ via the map:

$$n \mapsto [(n, 0)].$$

Check: If $m \neq n$, then $m + 0 \neq n + 0$, so $[(m, 0)] \neq [(n, 0)]$. Thus the map is injective, so we have no problem identifying $n \in \mathbb{N}$ with $[(n, 0)] \in \mathbb{Z}$.

[2] According to Leopold Kronecker, "Die ganzen Zahlen hat der liebe Gott gemacht, alles andere ist Menschenwerk" ("God made the integers, all else is the work of man"). Actually, here it is better to say that the integers are also the work of man, being formed out of the naturals.



Since we're thinking of the equivalence class of (a, b) to mean $a - b$, we can introduce standard terminology: positive if $a > b$, negative if $a < b$, and zero if $a = b$. Let us practice a bit with our definition to show a simple result, namely that this actually makes sense:

Proposition 2. *Every element of $\mathbb{Z}$ is well-defined to be either positive, negative, or zero.*

Proof. We need to show that the properties are well-defined on equivalence classes. Suppose $a + d = b + c$ and $a > b$, meaning there exists $k \in \mathbb{N}_+$ such that $a = b + k$. Then

$$b + k + d = b + c$$

so that $c = d + k$ with $k \in \mathbb{N}_+$, meaning (c, d) is also positive. The same argument works for the other cases. $\square$

Exercise 33. *Define subtraction on $\mathbb{Z}$.*

Arithmetic operations We had a notion of addition and multiplication for natural numbers, and we certainly expect we should have one for integers too. But we need to define it in terms of our equivalence classes now. For integers $[(a, b)]$ and $[(c, d)]$, define:

$$[(a, b)] + [(c, d)] := [(a + c,\ b + d)].$$

We need to check this is well-defined. If $(a, b) \sim_{\mathbb{Z}} (a', b')$ and $(c, d) \sim_{\mathbb{Z}} (c', d')$, then

$$(a + c,\ b + d) \sim_{\mathbb{Z}} (a' + c',\ b' + d').$$

Proof. Since $a + b' = b + a'$ and $c + d' = d + c'$, add them:

$$(a + b') + (c + d') = (b + a') + (d + c').$$

Rearrange:

$$(a+c)+(b'+d') = (b+d)+(a'+c').$$

This exactly means $(a+c,\ b+d) \sim_{\mathbb{Z}} (a'+c',\ b'+d')$. $\square$

Similarly, we define multiplication by

$$[(a,b)] \cdot [(c,d)] := [(ac+bd,\ ad+bc)].$$

Exercise 34. *Show this is well-defined.*

As we might expect from our intuition, the usual properties we showed hold for naturals also hold for integers:

Exercise 35. *Show that addition and multiplication of integers are commutative, associative, and satisfy distributivity.*

3.6 From $\mathbb{Z}$ to $\mathbb{Q}$

As we learned many years ago, there are situations in which even integers do not suffice to do mathematics, ranging from the mundane (splitting an apple), to the purely mathematical (we want to be able to solve equations like $1 = 2x$). Which one you find more motivating is a matter of taste, but the necessity of adding more numbers to our system becomes clear.

Just as we defined the integers out of the naturals, we will define rational numbers using a certain equivalence class on pairs of integers (now you may be starting to get a sense of what Kronecker meant). Let

$$Y = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) = \{(a,b) \mid a, b \in \mathbb{Z}, b \neq 0\}.$$

Define a relation $\sim_{\mathbb{Q}}$ on Y by:

$$(a,b) \sim_{\mathbb{Q}} (c,d) \iff a \cdot d = b \cdot c$$

where the $\cdot$ refers to multiplication as defined for $\mathbb{Z}$. Notice that this is none other than formalizing what we know about fractions as children!

Exercise 36. *Show that $\sim_{\mathbb{Q}}$ is an equivalence relation on Y.*

Definition 14 (Rational Numbers)**.** *The set of <u>rational numbers</u> is:*

$$\mathbb{Q} = Y/\sim_{\mathbb{Q}} = \{[(a,b)] \mid (a,b) \in Y\}$$

where $[(a,b)] = \{(c,d) \in Y \mid (a,b) \sim_{\mathbb{Q}} (c,d)\}$ is the equivalence class of (a,b).

We write $[(a,b)]$ as $\frac{a}{b}$, and use the standard notations like $0 = [(0,1)]$, $1 = [(1,1)]$, etc.

Once again, we want to be able to add and multiply rational numbers, but it is not immediately obvious how to do this. The solution, as before, is to simply codify what we already know about rational numbers in our hearts.

For addition, we define

$$[(a,b)] + [(c,d)] := [(ad+bc,\ bd)].$$

If $[(a,b)]$ represents $\frac{a}{b}$ and $[(c,d)]$ represents $\frac{c}{d}$, then:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}.$$

Thus $[(ad+bc,\ bd)]$ agrees with our intuition. We check it is well-defined:

Proof. Assume $(a,b) \sim_{\mathbb{Q}} (a',b')$ (so $ab' = ba'$) and $(c,d) \sim_{\mathbb{Q}} (c',d')$ (so $cd' = dc'$). We need to show:

$$(ad+bc,\ bd) \sim_{\mathbb{Q}} (a'd'+b'c',\ b'd').$$

That is:

$$(ad+bc)(b'd') = (bd)(a'd'+b'c').$$

From $ab' = ba'$ and $cd' = dc'$, compute:

$$\begin{aligned}(ad+bc)(b'd') &= adb'd' + bcb'd' \\ &= (ab')(dd') + (cd')(bb') \\ &= (ba')(dd') + (dc')(bb') \\ &= bd(a'd'+b'c').\end{aligned}$$

□

Likewise, we define multiplication:

$$[(a,b)] \cdot [(c,d)] := [(ac,\ bd)].$$

motivated by our usual intuition for multiplication of rationals.

Exercise 37. *Show that this is well-defined.*

Exercise 38. *Show that addition and multiplication of rationals are commutative, associative, and satisfy distributivity.*

The rational numbers are great, and they have a very nice feature which we will later refer to as forming a <u>field</u>: every nonzero element can be "inverted" (i.e. multiplied by something to get 1). This is extremely important for doing things like number theory. Let's prove it:

Proposition 3. *Given nonzero rational number r, there exists a unique rational number r^{-1} such that $r \cdot r^{-1} = 1 = [(1,1)]$.*

Proof. Let $q = [(a, b)] \neq 0$, so $a \neq 0$. Define $q' = [(b, a)]$ (valid since $a \neq 0$). Then

$$q \cdot q' = [(a, b)] \cdot [(b, a)] = [(ab, ba)].$$

Since integers commute, $ab = ba$, so $(ab, ba) \sim_{\mathbb{Q}} (1, 1)$ because $(ab) \cdot 1 = (ba) \cdot 1$. Hence

$$[(ab, ba)] = [(1, 1)] = 1.$$

Thus q' is a multiplicative inverse for q.

We need to check that it's unique. Assume $[(b, a)] \cdot [(a, b)] = 1$ and $[(b', a')] \cdot [(a, b)] = [(1, 1)]$, i.e., $ab' = a'b$. This is exactly the condition for $(b', a') \sim_{\mathbb{Q}} (b, a)$. Hence the inverse map is unique. □

Irrationality of $\sqrt{2}$ Nice as the rationals are, they are not enough for all purposes. In fact, thousands of years ago mathematicians were shocked to discover that[3] despite the equation containing nothing but natural numbers (not even having used integers or rationals!):

Theorem 1. *There is no rational solution to $x^2 = 2$.*

Before proving it, let us show:

Lemma 5. *If p is an integer and p^2 is even, then p is even.*

Proof. We prove the contrapositive: If p is odd, then p^2 is odd.
Let $p = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$p^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd. Hence, if p^2 is even, p cannot be odd; so p is even. □

We also need to introduce terminology you may be familiar with: for two integers $p, q \in \mathbb{Z}$, their <u>greatest common divisor</u> $\gcd(p, q)$ is the largest (positive) integer d such that $d|p$ and $d|q$. Now we proceed with the main result.

Proof of Theorem 1. Suppose there exists a rational number x such that $x^2 = 2$. Since x is rational, we can write

$$x = \frac{p}{q},$$

where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$ (i.e., the fraction is in lowest terms – this can always be done by dividing top and bottom a finite number of times[4]).

[3] In fact, some of the earliest known mathematics done by humans concerned this mysterious number "$\sqrt{2}$". Some of its digits were computed millenia ago in the Middle East, India, Africa, Europe, China, and probably elsewhere, before anything was known of irrational numbers.

[4] Proof: if $\gcd(p, q) = d > 1$, then $(p, q) \sim_{\mathbb{Q}} (\frac{p}{d}, \frac{q}{d})$ with $\frac{p}{d}, \frac{q}{d}$ both having fewer divisors (counted with multiplicity) than p, q. After doing this a finite number of times, there are no common divisors left except 1.

From $x^2 = 2$ we have

$$\left(\frac{p}{q}\right)^2 = 2 \quad \Rightarrow \quad \frac{p^2}{q^2} = 2 \quad \Rightarrow \quad p^2 = 2q^2. \tag{1}$$

Equation (1) tells us that $p^2 = 2q^2$, so p^2 is even. From the lemma, since p^2 is even, p is even. Thus we can write $p = 2k$ for some $k \in \mathbb{Z}$.

Substitute $p = 2k$ into equation (1):

$$(2k)^2 = 2q^2 \quad \Rightarrow \quad 4k^2 = 2q^2 \quad \Rightarrow \quad q^2 = 2k^2. \tag{2}$$

Equation (2) says $q^2 = 2k^2$, so q^2 is even. Applying the lemma again, q is even.

We have shown both p and q are even. This means $2 \mid p$ and $2 \mid q$, so $\gcd(p, q) \geq 2$. But this contradicts our initial choice that $\gcd(p, q) = 1$. $\square$

The purpose of the next chapter will be to remedy problems like this.

3.7 Constructing the natural numbers using Von Neumann's ordinals*

Peano's axioms tell us how to work with natural numbers. More precisely, they show that a set satisfying the axioms we can do the usual things like addition and multiplication, more generally, induction. We took the existence of such a set on faith, but strictly speaking, we haven't actually found such a set yet! Let's construct such an actual model $\mathcal{N}$ of the integers now, using set theory; it is not the only model, and you could work equivalently with others - the relationships between elements are what matters!

Construction First, we construct a set that we hope will be a model of the naturals. We'll first define certain sets, whose names will be given by numerals like "12" and "1098". We define

$$0 = \emptyset, \quad 1 = \{\emptyset\}, 2 = \{\emptyset, \{\emptyset\}\}, \quad 3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, ...$$

and so on. For each such set x, we define $S(x) = x \cup \{x\}$. In the rest of this section, we also denote $S(x) = x \cup \{x\}$ for <u>any</u> set x.

Definition 15. *We say a set X is inductive if $\emptyset \in X$ and $x \in X$ implies $S(x) \in X$.*

We'd like to construct an inductive set, but we can expect that such a set has to be "infinite" (whatever that means). What we've written down above is a bunch of finite sets of increasing size, but all of our tools so far are unable to produce an inductive set from these. We need to invoke the following axiom of set theory:

Axiom 3 (Axiom of Infinity)**.** *There exists an inductive set.*

Now take such an inductive set. By definition, this set contains

$$\emptyset, S(\emptyset), S(S(\emptyset)), S(S(S(\emptyset)))$$

and so forth. However, our it may also contain other stuff, which is surplus to requirements. We can get rid of this extra stuff, with the following definition:

Definition 16. *Let $\mathbb{N}$ be the smallest inductive set. More precisely:*

$$\mathbb{N} = \{x \mid x \in X \textit{ for all inductive sets } X\}.$$

This is our candidate model for the naturals.

Verifying the Peano axioms Let us now check that the set we constructed actually does what we want. We verify each condition:

- $0 \in \mathbb{N}$.

  By definition of $\mathbb{N}$ as intersection of all inductive sets, each such set contains 0, hence $0 \in \mathbb{N}$.

- Closure under successor: $\forall n \in \mathbb{N}, S(n) \in \mathbb{N}$.

  If $n \in \mathbb{N}$, then n belongs to every inductive set X. Each such X satisfies $x \in X \Rightarrow S(x) \in X$, thus $S(n) \in X$ for all such X. Hence $S(n) \in \mathbb{N}$.

- 0 is not a successor: $\forall n \in \mathbb{N}, S(n) \neq 0$.

  For any n, $S(n) = n \cup \{n\}$. Since $n \in S(n)$ but $n \notin \emptyset = 0$, we have $S(n) \neq 0$.

- Injectivity of successor:

  $$S(n) = S(m) \to n = m.$$

  This requires a lemma, given below. Suppose $S(n) = S(m)$, i.e.

  $$n \cup \{n\} = m \cup \{m\}.$$

  Since $n \in S(n)$, either $n \in m$ or $n = m$. Likewise, either $m \in n$ or $n = m$. If $n \neq m$, then $n \in m$ and $m \in n$. By the second statement in the lemma, we therefore have $n \in n$. But $n \subset n$, which contradicts the first statement of the lemma.

- Induction principle.

  Let P be a predicate such that:

  $$P(0) \quad \text{and} \quad \forall n \in \mathbb{N}\,(P(n) \to P(S(n))).$$

Define

$$A := \{n \in \mathbb{N} \mid P(n)\}.$$

Then $0 \in A$, and $n \in A \to S(n) \in A$, so A is an inductive set. Since $\mathbb{N}$ is the <u>smallest</u> inductive set, $\mathbb{N} \subset A$, but $A \subset \mathbb{N}$ by definition, so $A = \mathbb{N}$. Therefore $\forall n \in \mathbb{N}$, $P(n)$.

Lemma 6. *We have*

- *No natural number is a subset of any of its elements.*
- *Every element of a natural number is a subset of it.*

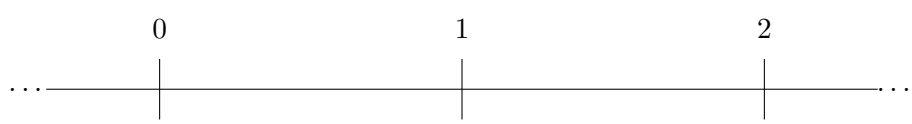
Proof. Optional exercise. □

Chapter 4

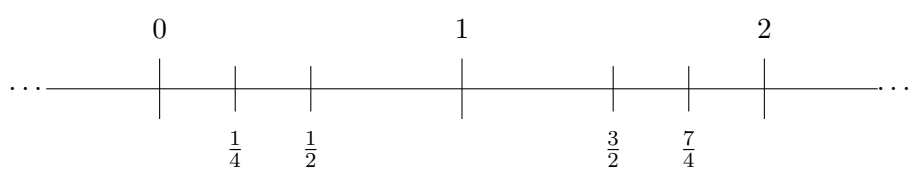
The reals

> Now when I was a little chap I had a passion for maps. I would look for hours at South America, or Africa, or Australia, and lose myself in all the glories of exploration. At that time there were many blank spaces on the earth, and when I saw one that looked particularly inviting on a map (but they all look that) I would put my finger on it and say, 'When I grow up I will go there.'
>
> Marlow, Heart of Darkness

Imagine an infinite ruler with integer spacings:



The rational numbers live between the integers:



As we add more and more rational numbers, our ruler gets more and more crowded.

No matter how many rationals we add, however, we can still find some blank spots. Our goal is to understand these mysterious blank spots: we call them the <u>real numbers</u>.

Unlike the integers or the rational numbers, calculations with real numbers are not daily features of human life. You may not have a clear idea of what real numbers really are. You might have seen heard:

- A real number is a point on the real line.
- A real number is something that looks like $\sqrt{2}$ or $\pi/3$.
- A real number is something that looks like

  $$10.234890234809234....$$

  In other words, it is given by a (potentially infinite) decimal expansion.

The first two answers are pretty vague. You may feel that the third answer is perfectly good: you already know how to add and multiply (potentially infinite) decimal expansions. However, there are a few problems with trying to define real numbers this way.

- A given real number may have many different decimal expansions. You may have learned in high school that

  $$0.999999.... = 1.$$

  As we shall see, one can fix this problem by considering decimal expansions up to a certain equivalence relation. But this redundancy suggests there might be a better, more direct definition.

- This strategy forces us to make the artificial choice of working in base 10, rather than for instance base 2 or base 3.

  We want a construction that does not force us to make this arbitrary choice.

- Perhaps the most serious problem with this strategy is that it does not clarify what is so useful and special about the real numbers.

In this chapter, we will answer the third point by writing down the basic properties which characterize the reals, and give a beautiful construction of them due to Dedekind.

The key to unlocking the reals is a structure we have not made much use of so far: the standard order $\leq$ on the rationals. We begin by studying it in more detail in the next section.

4.1 Total orders

Definition 17. *A set X with a partial order $\leq$ is* <u>*totally ordered*</u> *or* <u>*linearly ordered*</u> *if for all $x, y \in X$ either $x \leq y$ or $y \leq x$.*

For example, consider the usual $\leq$ relation on $\{1, 2, 3, 4\}$.

$$\begin{tikzcd}
4 \arrow[d, dash] \\
3 \arrow[d, dash] \\
2 \arrow[d, dash] \\
1
\end{tikzcd}$$

This is a total order, since any two elements can be compared.

For a non-example, consider $\{1, 2, 3, 6\}$ with the partial ordered defined by $x \leq y$ if x divides y. We can draw the Hasse diagram as follows:

$$\begin{tikzcd}
 & 6 \arrow[dl, dash] \arrow[dr, dash] & \\
2 \arrow[dr, dash] & & 3 \arrow[dl, dash] \\
 & 1 &
\end{tikzcd}$$

This is not a linear order, since 2 and 3 cannot be compared.

4.2 The standard order on $\mathbb{Z}$ and $\mathbb{Q}$

Previously, we defined subtraction for integers, and the notion of positivity and negativity. A similar construction can be made for rationals: we say $\frac{a}{b}$ is <u>positive</u> if the integers $a, b \in \mathbb{Z}$ are both positive or both negative, <u>negative</u> if one is positive and the other is negative, and <u>zero</u> if $a = 0$. We then define the standard order you are familiar with

$$x \leq y \text{ if } 0 \leq y - x$$

where we declare $0 \leq a$ if a is positive or zero. We can take $a \in \mathbb{Z}$ or $a \in \mathbb{Q}$, so that the standard order gives a partial order on $\mathbb{Z}$ or $\mathbb{Q}$.

Proposition 4. *The standard order $\leq$ on $\mathbb{Z}$ or $\mathbb{Q}$ is a total order.*

Proof. We have $x \leq y$ if and only if $0 \leq y - x$, and $x \geq y$ if and only if $0 \geq y - x$. Hence the claim follows for $\mathbb{Z}$ from the fact that every integer is either positive, negative or zero. The argument for rationals is similar. $\square$

Definition 18. *Let $(X, \leq_X)$ be a partially ordered set, and let $Y \subset X$ be a subset. We define the* <u>*restricted*</u> *partial order $\leq_Y$ on Y by*

$$y \leq_Y y' \leftrightarrow y \leq_X y'.$$

Usually, we drop the subscripts and simply write $\leq$ for both orders.

Exercise 39. *Suppose $(X, \leq)$ is totally ordered. Then any $Y \subset X$ equipped with the restricted order is also totally ordered.*

This gives us many examples of totally ordered sets, given by arbitrary subsets of the integers or rational numbers.

Of course, the rationals are not just an ordered set: they also have two binary operations $(+, \cdot)$. These play nicely with the order, as follows:

Proposition 5. *Let $x, y, z \in \mathbb{Q}$.*

- *If $x \leq y$, then $x + z \leq y + z$.*
- *If $x \geq 0$ and $y \geq 0$, then $x \cdot y \geq 0$.*

Exercise 40. *Prove it!*

4.3 Some fancy words

Let's take stock of what we know about $\mathbb{Z}$ and $\mathbb{Q}$, in order to see what is missing to define $\mathbb{R}$. This section is purely motivational.

Both $\mathbb{Z}$ and $\mathbb{Q}$ are sets equipped with distinguished elements $0, 1 \in \mathbb{Q}$ and binary operations $+, \cdot : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$. The pair of operations $(+, \cdot)$ are associative, commutative and distributive, and satisfy

$$0 + x = x \text{ and } 1 \cdot x = x \text{ for all } x.$$

Both $\mathbb{Z}$ and $\mathbb{Q}$ have additive inverses:

$$\text{For all } x, \text{ there exists } y \text{ such that } x + y = 0.$$

The rationals satisfy an additional property:

For all $x \neq 0$, there exists y such that $x \cdot y = 1$.

In other words, every element in $\mathbb{Q}$ has a multiplicative inverse.

Definition 19. *We call a set F equipped with elements $0, 1 \in F$ and binary operations $+, \cdot$ satisfying all the above properties a field.*

Hence, we can summarize almost everything we know so far about $\mathbb{Q}$ by saying '$\mathbb{Q}$ is a field'. Note, however, that the definition of field makes no mention of the order $\leq$ on $\mathbb{Q}$. Let's fix that.

Definition 20. *An ordered field is a field equipped with a total order $\leq$ satisfying:*

- *If $x \leq y$, then $x + z \leq y + z$.*
- *If $x \geq 0$ and $y \geq 0$, then $x \cdot y \geq 0$.*

By Proposition 5, $\mathbb{Q}$ is an ordered field. The reals will also be an ordered field. They will satisfy one extra property with respect to this order, which $\mathbb{Q}$ does not: completeness. The next few sections will build up to a definition of this property.

4.4 Upper bound, lower bound, infimum and supremum

In mathematics, interesting things tend to happen at the extremes - the biggest, the smallest, the first, the last. We need some words to describe these extremes.

Let Y be a partially ordered set.

- An element $y \in Y$ is a maximum for Y if for all $y' \in Y$, $y' \leq y$.
- An element $y \in Y$ is a minimum for Y if for all $y' \in Y$, $y' \geq y$.

Let X be a partially ordered set, and let $Y \subset X$ be a subset.

- An element $x \in X$ is an upper bound for Y if for all $y \in Y$, $y \leq x$.
- An element $x \in X$ is a lower bound for Y if for all $y \in Y$, $y \geq x$.

Let X be a partially ordered set, and let $Y \subset X$ be a subset.

- An upper bound $x \in X$ for Y is the least upper bound or supremum for Y if it minimal among upper bounds for Y, i.e. for any upper bound $x' \in X$ of Y, $x' \leq x$.

  We write $x = \sup(Y)$ (usually pronounced "soup").

- A lower bound $x \in X$ for Y is the <u>greatest lower bound</u> or <u>infimum</u> for Y if it is maximal among lower bounds for X, i.e. for any lower bound $x' \in X$ of Y, $x' \geq x$.

  We write $x = \inf(Y)$.

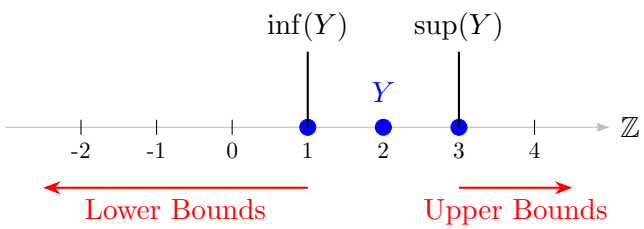


Figure 4.1: $X = \mathbb{Z}$ and $Y = \{1, 2, 3\}$.

Proposition 6. *A subset $Y \subset X$ has at most one supremum and at most one infimum.*

Proof. Suppose $x, x' \in X$ are two suprema[1] of Y. Since any supremum is also an upper bound, we have $x \geq x'$ and $x' \geq x$. By antisymmetry of $\geq$, we have $x = x'$.

The same argument holds for infima. □

Exercise 41. *Show that $\mathbb{Z}$ has no maximum or minimum.*

The set $\mathbb{Q}^+$ of (nonzero) positive rational numbers has no minimum, as the following proposition shows.

Proposition 7. *Let $q > 0$ be a positive rational number. Then there exists a rational number r such that $0 < r < q$.*

Proof. Pick $r = q/2$. We have

$$q - r = q - q/2 = q/2 > 0.$$

□

Note that the supremum and infimum of Y, when they exist, are not necessarily elements of Y. Consider the subset $\mathbb{Q}^+ \subset \mathbb{Q}$.

Proposition 8.

$$\inf(\mathbb{Q}^+) = 0.$$

Proof. Exercise. □

[1] We write suprema instead of supremums because supremum is a Latin word, and we won't let anyone forget it.

4.5 Completeness

Definition 21. *A partially ordered set X is* <u>*complete*</u> *if every* ***non-empty*** *subset $Y \subset X$ with an upper bound has a supremum, and every* ***non-empty*** *set $Y \subset X$ with a lower bound has an infimum.*

Example 16. *Any finite ordered set is complete.*

Proposition 9. *The ordered set $(\mathbb{Q}, \leq)$ of rational numbers is not complete.*

Proof. We saw in the previous chapter that there is no rational solution to the equation $p^2 = 2$. Consider the set

$$S = \{x \in \mathbb{Q} \mid x^2 < 2\}.$$

We claim that S has an upper bound, but does not have a supremum. The next three exercises conclude the proof. □

Exercise 42. *Show that every s such that $s^2 > 2$ is an upper bound for S.*

Exercise 43. *Suppose $s^2 < 2$. Then there exists $r \in \mathbb{Q}$, $s < r$ such that $r^2 < 2$. Hence s is not an upper bound for S.*

Hint: Write $M = 2 - s^2$. It is enough to show that there exists $\epsilon > 0$ such that $\epsilon(2s + \epsilon) < M$.

By Proposition 7, there exists $0 < \epsilon < 2s$. In this case,

$$\epsilon(2s + \epsilon) < \epsilon(4s).$$

Thus it is enough to find ϵ such that $\epsilon < 2s$ and

$$\epsilon < \frac{M}{4s}.$$

A second application of Proposition 7 finishes the proof.

Exercise 44. *Suppose $s^2 > 2$. Then there exists $r \in \mathbb{Q}$, $r < s$ which is an upper bound for S. Hence s is not a supremum of S.*

4.6 Axioms for the real numbers

When we defined the natural numbers, we started from a collection of axioms (the Peano axioms) which tell us what we want from the naturals. Only later did we construct a set satisfying these axioms. We now do something similar for the real numbers.

Definition 22 (Formal definition)**.** *We say a set R is a model of the real numbers if it is equipped with:*

- *Elements* $0, 1 \in R$,
- *Binary operations* $+, \cdot : R \times R \to R$,
- *A total order* $\leq$,

such that

- $(R, 0, 1, +, \cdot, \leq)$ *is an ordered field.*
- *The ordered set* $(R, \leq)$ *is complete.*

4.7 A first attempt at constructing the reals

Now that we know what properties we want the reals to have, we can begin to construct them. We will use the informal definition in terms of decimal expansions as a guide. Suppose we are given a real number as a decimal expansion:

$$r = 10.234890234809234....$$

We can <u>approximate</u> r with better and better accuracy by consider more and more digits in the expansion:

$$10 < r < 11$$
$$10.2 < r < 10.3$$
$$10.23 < r < 10.24$$
$$...$$

Our basic strategy will be to turn these approximations into a definition. We will think of r as dividing the rational numbers into two camps:

$$A = \{q \in \mathbb{Q} \mid q < r\}$$

and

$$B = \{q \in \mathbb{Q} \mid q > r\}.$$

The key idea is to define r as the unique solution to

$$A < r < B.$$

We will see how to do this in the next section.

4.8 Defining the reals using Dedekind cuts

Definition 23. *A* <u>*Dedekind cut*</u> *is a pair* (A, B) *of nonempty subsets of* $\mathbb{Q}$ *such that:*

- $\mathbb{Q} = A \cup B$
- *For all* $a \in A$ *and* $b \in B$, $a < b$.
- A *does not have a* ***maximum****.*

Definition 24. *(Equivalent definition of Dedekind cuts) A subset* $A \subset \mathbb{Q}$ *is called a* <u>*Dedekind cut*</u>*, if*

1. $A \neq \emptyset$, $A \neq \mathbb{Q}$,
2. *For every* $x \in A$ *and every* $y \in \mathbb{Q}$ *such that* $y < x$*, we have* $y \in A$,
3. *For every* $x \in A$ *there exists* $y \in \mathbb{Q}$ *such that* $x < y$ *and* $y \in A$.

Exercise 45. *Let* (A, B) *is a Dedekind cut in the sense of Definition 23. Prove that* A *is a Dedekind cut in the sense of Definition 24.*

Conversely, let A *be a Dedekind cut in the sense of Definition 24. Prove that* $(A, \mathbb{Q} \setminus A)$ *is a Dedekind cut in the sense of Definition 23.*

Definition 25. *Let* $\mathbb{R}$ *be the set of Dedekind cuts.*

When we write $(A, B) \in \mathbb{R}$ or say that (A, B) is a Dedekind cut, we mean it in the sense of Definition 23. On the other hand, if only one set A is explicitly mentioned, we imply Definition 24.

We now need to define the structures $0, 1, +, \times, \leq$ and show they satisfy the requisite properties.

First, however, we will show that our definition of the real numbers contains a copy of the rationals.

Definition 26. *We say a Dedekind cut* (A, B) *is* <u>*rational*</u> *if* B *has a minimum.*

Proposition 10. *The function*

$$(A, B) \to \min(B)$$

is a bijection between rational Dedekind cuts and rational numbers.

Proof. □

We will denote a rational Dedekind cut (A, B) by the rational number $\min(B) \in \mathbb{Q}$ whenever convenient. In particular, we have the distinguished elements

$$0, 1 \in \mathbb{R}$$

corresponding to $0, 1 \in \mathbb{Q}$.

For a rational number $q \in \mathbb{Q}$, we define $q^{\mathbb{R}}$ to be the Dedekind cut $\{x \in \mathbb{Q} : x < q\}$.

Exercise 46. *Show that for each $q \in \mathbb{Q}$, the set $q^{\mathbb{R}}$ is indeed a Dedekind cut. Moreover, show that the function $i : \mathbb{Q} \to \mathbb{R}$ such that $i(q) = q^{\mathbb{R}}$ is an injection.*

Remark 6. *For a rational number $q \in \mathbb{Q}$, show that the set $\{x \in \mathbb{Q} : x \leq q\}$ satisfies the conditions 1 and 2 of Definition 24, but not 3. This explains the importance of the condition 3 – it helps to avoid ambiguity of embedding of $\mathbb{Q}$ into $\mathbb{R}$.*

Next, let's construct the operations of addition and multiplication on Dedekind cuts. Although the following constructions may seem a little surprising or ad-hoc at first, once you play around with them a bit you will find they are quite natural.

Definition 27. *Given Dedekind cuts A and A', define*

$$A + A' = \{a + a' \mid a \in A, a' \in A'\} \tag{4.1}$$

Proposition 11. *The right-hand side of (4.1) is a Dedekind cut. If A, A' are rational, then the sum defined as above coincides with the usual addition of rational numbers.*

Proof. Let us check that $A + A'$ satisfies the conditions 1,2,3 from Definition 24.

1. By Definition 24.1 applied to A and A', we have $A \neq \emptyset$ and $A' \neq \emptyset$. Therefore, we can find elements $a \in A$ and $a' \in A'$. This gives an element $a+a'$ of $A+A'$. This shows that $A + A' \neq \emptyset$.

   By Definition 24.1 applied to A and A', we have $A \neq \mathbb{Q}$ and $A' \neq \mathbb{Q}$. Therefore, we can find elements $b \in \mathbb{Q} \setminus A$ and $b' \in \mathbb{Q} \setminus A'$. Let us show that $b+b' \notin A+A'$. To seek a contradiction, let $b + b' = a + a'$, for some $a \in A$, $a' \in A'$. By Definition 24.2 applied to A, we obtain $\neg(b < a)$, i.e. $a \leq b$. Additionally, since $a \in A$, $b \notin A$, we have $a \neq b$. Therefore, $a < b$. Same argument applied to primed version A' shows $a' < b'$. Thus we have $a + a' < b + a' < b + b'$, which contradicts the earlier established equality $a+a' = b+b'$. Thus we have proved $b + b' \notin A + A'$, and so $A + A' \neq \mathbb{Q}$.

2. Let $x \in A + A'$, and $y \in \mathbb{Q}$ be such that $y < x$. We need to show that $y \in A + A'$.

   By definition of $A + A'$, we have $x = a + a'$ for some $a \in A$, $a' \in A'$. By Definition 24.2 applied to A, we obtain that $y - a' < a$, $a \in A$ imply that $y - a' \in A$. This in turn implies that $y \in A + A'$, as desired.

3. Let $x \in A + A'$. Let present $y \in A + A'$ such that $x < y$.

   Let $x = a + a'$, for some $a \in A$, $a' \in A'$. By Definition 24.3 applied to A and A', we can find $b \in A$ with $a < b$ and $b' \in A'$ with $a' < b'$. Then $y = b + b'$ is the desired element of $A + A'$. Indeed, we have

$$x = a + a' < b + a' < b + b' = y.$$

We leave it as an exercise to check that $A \in \mathbb{Q}$, $A' \in \mathbb{Q}$ imply $A + A' \in \mathbb{Q}$. $\square$

Definition 28. *We define a partial order on $\mathbb{R}$ by*

$$(A, B) \leq (A', B')$$

if $A \cap B' = \emptyset$. Equivalently, in terms of Definition 24, we define $A \leq A'$ if A is a subset of A'. Notation $A < A'$ means that $A \leq A'$ and $A \neq A'$.

Proposition 12. *The above relation $\leq$ is (as-claimed) a partial order. It extends the previously defined partial order on $\mathbb{Q}$.*

Proposition 13. *The reals are totally ordered.*

Proof. Exercise. $\square$

For a Dedekind cut $A \in \mathbb{R}$, we define $-A$ to be the unique Dedekind cut such that $(-A) + A = 0^{\mathbb{R}}$.

Exercise 47. *If $A = q^{\mathbb{R}}$ for some $q \in \mathbb{Q}$, then $-A = (-q)^{\mathbb{R}}$.*
If $A \in \mathbb{R}$ is not rational, then $-A = \mathbb{Q} \setminus \{x \in \mathbb{Q} \mid -x \in A\}$.

Definition 29. *Let A, A' be two Dedekind cuts.*
If $0^{\mathbb{R}} \leq A$ and $0^{\mathbb{R}} \leq A'$, we define

$$A \cdot A' = \cancel{\{x \in \mathbb{Q} \mid \forall b \in \mathbb{Q} \setminus A, \forall b' \in \mathbb{Q} \setminus A' : x < b \cdot b'\}} =$$

$$= \{a \cdot a' | a \in A, a' \in A', a \geq 0, a' \geq 0\} \cup \{x \in \mathbb{Q} | x < 0\}.$$

If $0^{\mathbb{R}} \geq A$ and $0^{\mathbb{R}} \leq A'$, we define

$$A \cdot A' = A' \cdot A = -((-A) \cdot A').$$

If $0^{\mathbb{R}} \geq A$ and $0^{\mathbb{R}} \geq A'$, we define

$$A \cdot A' = (-A) \cdot (-A').$$

Proposition 14. *The operations $(+, \cdot)$ make $\mathbb{R}$ into a field. They restrict to the previously defined $(+, \cdot)$ operations on $\mathbb{Q} \subset \mathbb{R}$.*

Proof. There are many things to check here! I recommend you try verifying one or two of the properties to get a general idea of how the proofs go. Note, however, that essentially all of the proofs piggy-back on the corresponding properties of the rational numbers. $\square$

Proposition 15. *The reals are an ordered field.*

Proof. Exercise. $\square$

Proposition 16. *The reals are complete.*

Exercise 48. *Prove the proposition.*

4.9 Uniqueness of the reals

The Dedekind cut construction may seem a tiny bit arbitrary. Sure, it reproduces many familiar and expected features of the real numbers. But why this particular construction, and not another?

It is a wonderful theorem, which unfortunately we cannot prove in this class, that there is a unique complete ordered field.[2] Hence any other construction will yield the same result!

4.10 Decimal expansions

How does the Dedekind-cut definition compare with our familiar decimal expansions?

Consider

$$x = 10.4352902543....$$

We would like to associate to this decimal expansion a Dedekind cut. We consider the following sequence of lower bounds for x:

$$S = \{10, 10.4, 10.43, 10.435, ...\}$$

Define:

$$A = \{q \in \mathbb{Q} \mid \exists s \in S \text{ such that } q < s\}.$$

Now consider the following sequence of upper bounds for x:

$$T = \{11, 10.5, 10.44, 10.436, ...\}$$

Define:

$$B = \{q \in \mathbb{Q} \mid \exists t \in T \text{ such that } q > t).$$

We claim that (A, B) are a Dedekind cut. This cut is precisely the real number defined by the decimal expansion.

Exercise 49. *Given a Dedekind cut, how would you produce a decimal expansion for the corresponding real number?*

4.11 Irrationals

Definition 30. *An irrational number is an element $r \in \mathbb{R} \setminus \mathbb{Q}$.*

Exercise 50. *Show that there exists a positive real number $r \in \mathbb{R}$ such that $r^2 = 2$.*

$$A = \{r \in \mathbb{Q} \mid r \leq 0 \text{ or } r^2 < 2\}$$
$$B = \mathbb{Q} \setminus A = \{r \in \mathbb{Q} \mid r > 0 \text{ and } r^2 > 2\}$$

We call this number $\sqrt{2}$; it is an example of an irrational number.

[2] Later in this class, we will say more about what we mean by uniqueness in mathematics.

4.12 Sequences and limits

Definition 31. *Let X be a set. A* <u>*sequence*</u> *in X is a function*

$$\mathbb{N} \to X.$$

We denote sequences by

$$(x_n)$$

where x_i is the image of $i \in \mathbb{N}$.

As a special case of the above:

Definition 32. *A sequence of rational numbers is a map $\mathbb{N} \to \mathbb{Q}$. A sequence of real numbers is a map $\mathbb{N} \to \mathbb{R}$.*

Sequences of integers or rational numbers are probably familiar concepts to you: you may have worked with sequences such as

$$1, 2, 3, 4, ... \tag{4.2}$$

or

$$1, 4, 9, 16, ... \tag{4.3}$$

or

$$1, 1/2, 1/3, 1/4, ... \tag{4.4}$$

where the ... invite you to guess the remaining terms. However, in general there is no requirement that the terms (x_n) of a sequence be governed by some simple formula.

Some sequences, like the first two above, seem to wander off to infinity. Others, like the third sequence above, get closer and closer to some known destination (zero, in this case).

Definition 33. *A sequence (x_n) of real numbers has limit $L \in \mathbb{R}$ if for all positive real numbers $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$,*

$$|L - x_n| < \epsilon.$$

We write

$$L = \lim_{n \to \infty} x_n.$$

This definition is a mouthful! If a sequence (x_n) has a limit L, we sometimes also say that the sequence <u>converges to L</u>.

Exercise 51. *Write this out more symbolically.*

Example 17. *The sequence* (4.4) *has limit $L = 0$.*

Exercise 52. *Compute the limit (if it exists) of*

$$x_n = 0 \textit{ for all} n \geq 0$$

Example 18. *Let*

$$x_n = \frac{5n+2}{3n-7}, n \geq 0.$$

We show that $\lim_{n\to\infty} x_n = \frac{5}{3}$.

$$\begin{aligned}
\left|\frac{5n+2}{3n-7} - \frac{5}{3}\right| &= \left|\frac{41}{3(3n-7)}\right| \\
&= \frac{41}{3(3n-7)} \\
&\leq \frac{41}{3(2n)} \quad \textit{(choose } N \textit{ s.t. } 3n-7 \geq 2n\textit{)} \\
&= \frac{41}{6n} \\
&< \frac{41}{6 \cdot \frac{41}{6\epsilon}} \quad \textit{(choose } N > \frac{41}{6\epsilon}\textit{)} \\
&= \epsilon.
\end{aligned}$$

4.13 Cauchy sequences*

In this section, we discuss an alternative construction of the real numbers. So, let's pretend we never learned about Dedekind cuts, and start again from the rationals.

Instead of Dedekind cuts, the key concept here is to represent real numbers by sequences of rational numbers whose terms get closer and closer together in a certain way (note that we certainly need some kind of restriction, since an arbitrary sequence could do lots of unpleasant things).

Here is a sequence that seems to be getting closer and closer to something, without us being able to name that something:

$$1 \tag{4.5}$$

$$1 + \frac{1}{2}, \tag{4.6}$$

$$1 + \frac{1}{2} + \frac{1}{3 \cdot 2} \tag{4.7}$$

$$1 + \frac{1}{2} + \frac{1}{3 \cdot 2} + \frac{1}{4 \cdot 3 \cdot 2 \cdot 1} \tag{4.8}$$

$$.... \tag{4.9}$$

If you draw the successive terms of this sequence, you will find that they seem to closer and closer to a limiting point L on the line. However, try as you might, you will not find a rational number which is the limit of this sequence.

Definition 34. *A sequence (x_n) is* <u>*Cauchy*</u> *if for all $\epsilon > 0$, there exists N such that $|x_n - x_m| < \epsilon$ for all $n, m > N$.*

Exercise 53. *Find some examples of Cauchy sequences, and sequences which fail to be Cauchy.*

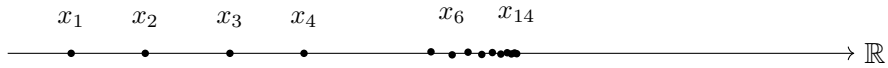
What this definition tries to capture is the idea that the terms of the sequence are getting closer and closer to some limiting point L on the line. This limit point will be - you guessed it - a real number.

We might want to say: $|L - x_n|$ should get smaller and smaller as n gets bigger and bigger, i.e.

$$\lim x_n = L.$$

However, since we are trying to <u>define</u> real numbers using Cauchy sequences, we cannot yet speak directly of L.

Instead, we focus on the difference between terms x_n and x_m as n, m get larger and larger. We ask that as n gets bigger and bigger, the x_n wander around in a smaller and smaller zone.



Proposition 17. *Suppose the sequence (x_n) has a limit $L \in \mathbb{Q}$. Then (x_n) is a Cauchy sequence.*

Proof. Difficult exercise. □

Definition 35. *Two Cauchy sequences (x_n) and (y_n) are* <u>*equivalent*</u> *if for all $\epsilon > 0$, there exists N such that $|x_n - y_m| < \epsilon$ for all $m, n > N$.*

The idea here is that two equivalent Cauchy sequences are limiting to the same point on the line.

Proposition 18. *The above definition yields an equivalence relation on the set of Cauchy sequences.*

Proof. Exercise. You will need the triangle inequality. □

Proposition 19 (Triangle inequality for rationals)**.** *For any two rational numbers $a, b \in \mathbb{Q}$, we have*

$$|a + b| \leq |a| + |b|$$

Proof. Exercise. □

Definition 36. *We write $\mathbb{R}$ for the set of equivalence classes of Cauchy sequences of rational numbers, and call it the* <u>*real numbers*</u> *or simply the* <u>*reals*</u>*.*

Exercise 54. *How can we think of rationals in this way?*

Exercise 55. *Find a few representatives of $\frac{1}{2} \in \mathbb{Q}$*

We now have a set - it remains to define operations on it! Much as with the definition via Dedekind cuts, our construction of operations will piggy-back on the corresponding operations for the rational numbers. We take the following lemma for granted (you may prove it as an exercise):

Lemma 7. *Every Cauchy sequence is bounded.*

Proposition 20. *If (x_n) and (y_n) are Cauchy sequences, then so are $(x_n + y_n)$ and $(x_n y_n)$.*

Proof. The case of addition is easier and left as an exercise. For multiplication, using the triangle inequality we have:

$$\begin{aligned}
|x_m y_m - x_n y_n| &= |x_m y_m - x_m y_n + x_m y_n - x_n y_n| \\
&\leq |x_m y_m - x_m y_n| + |x_m y_n - x_n y_n| \\
&= |x_m| \cdot |y_m - y_n| + |y_n| \cdot |x_m - x_n| \\
&\leq M|y_m - y_n| + M|x_m - x_n|.
\end{aligned}$$

where we used the boundedness to obtain a common bound M for (x_n) and (y_n) in the last step. Given $\varepsilon > 0$:

- Since (y_n) is Cauchy, $\exists N_1 \in \mathbb{N}$ such that $\forall m, n \geq N_1$,
$$|y_m - y_n| < \frac{\varepsilon}{2M}.$$
- Since (x_n) is Cauchy, $\exists N_2 \in \mathbb{N}$ such that $\forall m, n \geq N_2$,
$$|x_m - x_n| < \frac{\varepsilon}{2M}.$$

Let $N = \max(N_1, N_2)$. Then for all $m, n \geq N$:

$$|x_m y_m - x_n y_n| \leq M \cdot \frac{\varepsilon}{2M} + M \cdot \frac{\varepsilon}{2M} = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $(x_n y_n)$ is a Cauchy sequence in $\mathbb{Q}$. $\square$

Proposition 21. *Sums and products are well-defined on equivalence classes, and therefore define operations*

$$+ : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$

and

$$\cdot : \mathbb{R} \times \mathbb{R} \to \mathbb{R}.$$

Proof. Exercise. $\square$

Exercise 56. *Show that every nonzero real number has a multiplicative inverse, i.e. given* $[(x_n)] \neq [(0, 0, 0, \ldots)]$*, there is a* $[(y_n)]$ *such that*

$$[(x_n y_n)] = [(1, 1, 1, \ldots)]$$

Definition 37. *Let* x, y *be real numbers. We write* $x \leq y$ *if there exists* $N \in \mathbb{N}$ *such that for all* $n, m > N$*,*

$$x_n \leq y_n.$$

Theorem 2. *The reals are a complete ordered field.*

Proof. Exercise. $\square$

Chapter 5

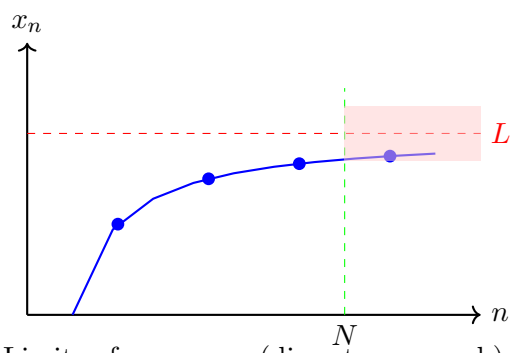
Delta and Epsilon

5.1 From Sequences to Functions: Why We Need Delta-Epsilon

5.1.1 Recap: Limits of Sequences

In the previous chapter we discussed limits of sequences:

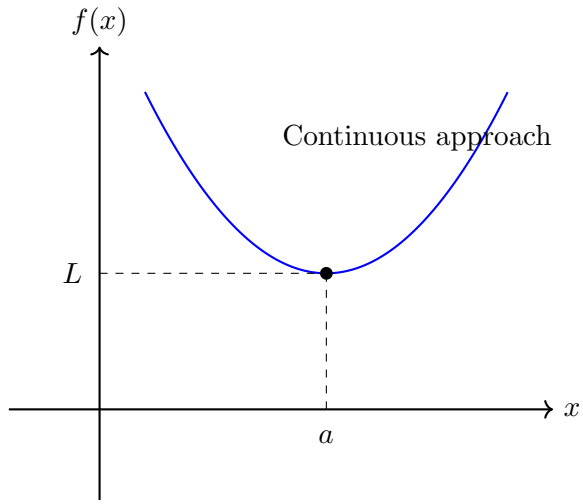
$$\lim_{n\to\infty} x_n = L \iff \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } n \geq N \implies |x_n - L| < \varepsilon$$



Limits of sequences (discrete approach)

That is, we can make x_n arbitrarily close to L by going far enough out in the sequence. We would like to now consider functions not only like $x : \mathbb{N} \to \mathbb{R}$, but $f : U \to \mathbb{R}$ from more general domains U.

In essence, we want to capture: as x approaches a, $f(x)$ approaches L. The new difficulty is that x approaches a "continuously" from both sides, not just in discrete steps.



5.2 The Formal Definition and First Examples

5.2.1 The Formal Definition

First, we need to be a bit more careful about what kind of functions we will consider. We want to consider functions with a "continuous" domain – for example, considering

$$f : \mathbb{Q} \to \mathbb{R}$$

although technically possible, is a bit unpleasant to think about. The purpose of inventing the reals, after all, was to have a number system without so many holes in it.

Definition 38. *An* <u>*open interval*</u> *is a subset $I \subset \mathbb{R}$ of the form*

$$(b, c) := \{x \in \mathbb{R} \mid b < x < c\}$$

or

$$(-\infty, c) := \{x \in \mathbb{R} \mid x < c\}$$
$$(b, \infty) := \{x \in \mathbb{R} \mid b < x\}$$

where $b, c \in \mathbb{R}$ and $b < c$.

To discuss limits of functions, we will consider functions defined on open intervals with a point removed.

Definition 39. *Suppose I is an open interval containing $a \in \mathbb{R}$. Let $f : I \setminus \{a\} \to \mathbb{R}$ be defined on the complement of $\{a\}$. We say:*

$$\lim_{x \to a} f(x) = L$$

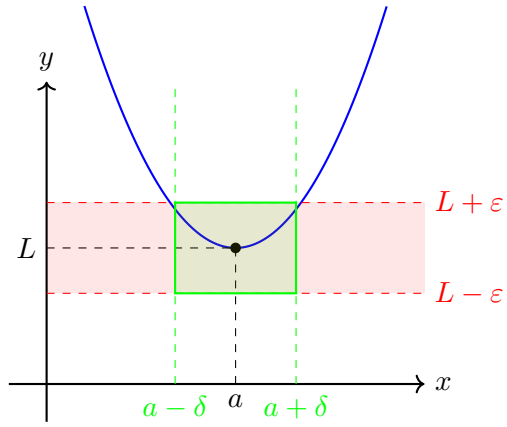


Figure 5.1: For every $\epsilon > 0$, if x is in a small enough green interval then $f(x)$ is in the red band

if and only if:

$$\boxed{\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x-a| < \delta \implies |f(x) - L| < \varepsilon}$$

When the domain is obvious, or just a small enough interval centred on a point a, we might write the explicit functional form of $f(x)$ without specifying the precise domain (see the examples below).

5.2.2 Examples of Limits

Example 19. *Let us compute*

$$\lim_{x \to 3} (2x+1) = 7.$$

Let $f(x) = 2x+1$, $a = 3$, $L = 7$. *We check*

$$|f(x) - 7| = |(2x+1) - 7| = |2x - 6| = 2|x-3|$$

We want $2|x-3| < \varepsilon$, *which means* $|x-3| < \frac{\varepsilon}{2}$. *We choose*

$$\delta = \frac{\varepsilon}{2}.$$

Thus, given $\varepsilon > 0$, *let* $\delta = \frac{\varepsilon}{2}$. *If* $0 < |x-3| < \delta$, *then:*

$$|f(x) - 7| = 2|x-3| < 2\delta = 2 \cdot \frac{\varepsilon}{2} = \varepsilon$$

Thus $\lim_{x \to 3}(2x+1) = 7$.

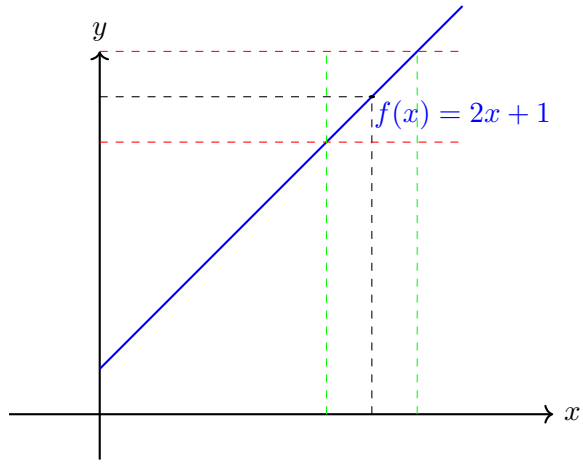


Figure 5.2: $\delta = \frac{\varepsilon}{2}$ works for any ε

Exercise 57. *Suppose we already know the fact that* $\lim_{x\to -2}(4x+3) = -5$. *Let us find a* δ *(guaranteed to exist) for* $\varepsilon = 0.1$.

We compute $|f(x) - L| = |(4x+3) - (-5)| = |4x+8| = 4|x+2|$. *We would like* $4|x+2| < 0.1 \implies |x+2| < 0.025$, *which will clearly be satisfied if* $\delta = 0.025$.

Example 20 (Constant functions)**.** *We certainly expect that for a constant function* $f \equiv c$ *that* $\lim_{x\to a} c = c$. *Let's show this. For any* a *and constant* c, *given* $\varepsilon > 0$, *<u>any</u>* $\delta > 0$ *works because:*

$$|f(x) - c| = |c - c| = 0 < \varepsilon \quad \textit{for all } x$$

So we can pick $\delta = 1$, *or* $\delta = \varepsilon$, *or anything!*

Example 21 (A Function with a Hole)**.** *Let's consider*

$$f : (0, \infty) \setminus \{2\} \to \mathbb{R}$$

given by $f(x) = \frac{x^2-4}{x-2}$. *Let us show that* $\lim_{x\to 2} \frac{x^2-4}{x-2} = 4$. *Note:* $f(x) = \frac{x^2-4}{x-2}$ *is undefined at* $x = 2$, *but limits don't care!*

Note that for $x \neq 2$

$$\frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x + 2.$$

So near $x = 2$, $f(x) = x + 2$.

$$|f(x) - 4| = |(x+2) - 4| = |x - 2|$$

We want $|x - 2| < \varepsilon$, *so choose* $\delta = \varepsilon$.

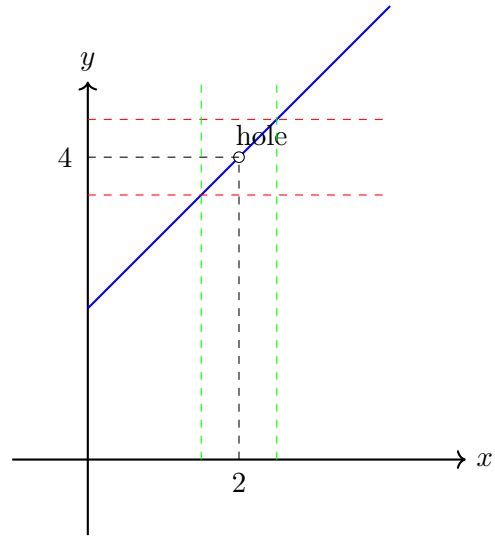


Figure 5.3: Even with a hole, the limit exists!

Thus given $\varepsilon > 0$, let $\delta = \varepsilon$. If $0 < |x-2| < \delta$, then:

$$|f(x) - 4| = |x - 2| < \delta = \varepsilon$$

Thus the limit exists and equals 4, even though $f(2)$ is undefined!

Exercise 58. *Prove* $\lim_{x\to 4} \sqrt{x} = 2$.

5.3 When Limits Don't Exist

So far the examples we gave were quite nicely behaved - we successfully found a limit each time. This certainly does not have to be the case, and we will now see a few ways in which it can fail.

For any $L \in \mathbb{R}$, to show $\lim_{x\to a} f(x) \neq L$, we need to find <u>one</u> $\varepsilon > 0$ such that for <u>every</u> $\delta > 0$, there exists some x with $0 < |x-a| < \delta$ but $|f(x) - L| \geq \varepsilon$. That is,

$$\lim_{x\to a} f(x) \neq L \iff \exists \varepsilon > 0 \text{ such that } \forall \delta > 0, \exists x \text{ with } 0 < |x-a| < \delta \text{ and } |f(x) - L| \geq \varepsilon$$

Example 22. *Let $f(x) = \frac{|x|}{x}$. Note:*

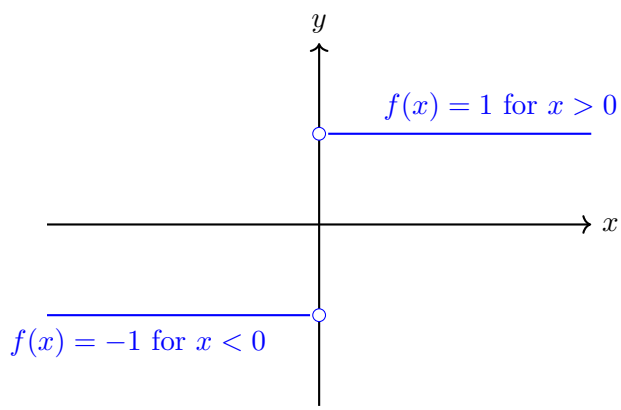
$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

We claim that $\lim_{x\to 0} \frac{|x|}{x}$ does not exist. To prove this, choose $\varepsilon = 1$. For any $\delta > 0$:

- *Pick* $x_1 = \delta/2 > 0$. *Then* $|x_1 - 0| < \delta$ *and* $f(x_1) = 1$
- *Pick* $x_2 = -\delta/2 < 0$. *Then* $|x_2 - 0| < \delta$ *and* $f(x_2) = -1$

No single L *can be within* $\varepsilon = 1$ *of both 1 and -1 simultaneously.*

Thus the limit does not exist.

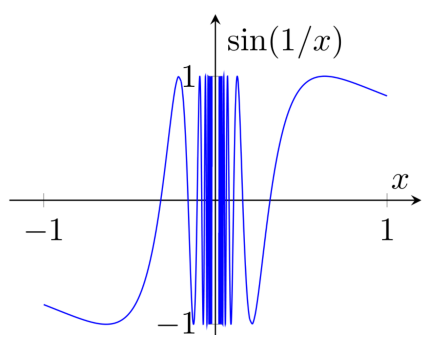


Exercise 59. *Let* $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ *given by*

$$f(x) = \sin(1/x)$$

for $x > 0$. *Show that* $\lim_{x\to 0} f(x)$ *does not exist.*

We can see that the value oscillates forever between 0 and 1 (or -1 and 1) arbitrarily close to 0.



5.4 Algebra of Limits

In the previous sections, we proved various limits from scratch, but it took some effort. In favourable circumstances, we can make things easier for ourselves by combining previously known limits to calculate new ones.

Proposition 22. *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$*, then:*

$$\lim_{x\to a}[f(x) + g(x)] = L + M$$

Proof.* Given $\varepsilon > 0$, we need $\delta > 0$ such that $|f(x) + g(x) - (L + M)| < \varepsilon$.

Note by the triangle inequality, we have $|f(x) + g(x) - (L + M)| \leq |f(x) - L| + |g(x) - M|$.

Since $\lim f = L$, for $\varepsilon/2$, we can find $\delta_1 > 0$ such that $|x - a| < \delta_1 \implies |f(x) - L| < \varepsilon/2$.

Likewise, since $\lim g = M$, for $\varepsilon/2$, we can find $\delta_2 > 0$ such that $|x - a| < \delta_2 \implies |g(x) - M| < \varepsilon/2$.

Then we choose $\delta = \min(\delta_1, \delta_2)$. If $|x - a| < \delta$, then:

$$|f(x) + g(x) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

□

Example 23. *Let* $f, g : (0, \infty) :\to \mathbb{R}$ *with* $f(x) = 2x + 1$*,* $g(x) = \sqrt{x}$*. Then since both limits exist (following previous examples and exercises)*

$$lim_{x\to 4}(f(x) + g(x)) = \lim_{x\to 4} f(x) + \lim_{x\to 4} g(x) = 9 + 2 = 11$$

Exercise 60. *Find a pair of functions for which* $\lim_{x\to a} f(x)$ *and* $\lim_{x\to a} g(x)$ *do not exist, but* $\lim_{x\to a}(f(x) + g(x))$ *exists.*

Likewise, for products we have

Proposition 23. *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$*, then:*

$$\lim_{x\to a}[f(x)g(x)] = LM$$

If $M \neq 0$

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}$$

Proof.*

$$\begin{aligned} |f(x)g(x) - LM| &= |f(x)g(x) - f(x)M + f(x)M - LM| \\ &\leq |f(x)||g(x) - M| + |M||f(x) - L| \end{aligned}$$

We need to bound $|f(x)|$ near a. Since $\lim f = L$, pick δ_1 so that $|x - a| < \delta_1 \implies |f(x) - L| < 1$, so $|f(x)| < |L| + 1$.

Then for given $\varepsilon > 0$, choose:

- δ_1 so $|f(x)| < |L| + 1$

- δ_2 so $|g(x) - M| < \frac{\varepsilon}{2(|L|+1)}$
- δ_3 so $|f(x) - L| < \frac{\varepsilon}{2|M|}$ (if $M \neq 0$)

Now choose $\delta = \min(\delta_1, \delta_2, \delta_3)$. Then:

$$|f(x)g(x) - LM| < (|L| + 1)\frac{\varepsilon}{2(|L| + 1)} + |M|\frac{\varepsilon}{2|M|} = \varepsilon$$

For quotients, we leave the proof as an exercise. □

Example 24. *Let us compute* $\lim_{x\to 0} \frac{x^2-4}{x-1}$ *using limit laws. Since the limit of the denominator is not zero, we can write*

$$\lim_{x\to 0} \frac{x^2 - 4}{x - 1} = \lim_{x\to 0} \frac{(x - 2)(x + 2)}{x - 1} = \frac{\lim_{x\to 0}(x - 1) \lim_{x\to 0}(x + 2)}{\lim_{x\to 0}(x - 1)} = \frac{-2 \cdot 2}{-1} = 4$$

Thus $\lim_{x\to 0} \frac{x^2-4}{x-2} = 4$

Exercise 61. *Try computing this limit without using limit laws.*

5.5 Connection to Sequence Limits

Theorem 3 (Sequential Characterization)**.** *Let* $f : I \setminus \{a\} \to \mathbb{R}$ *with*

$$\lim_{x\to a} f(x) = L$$

if and only if for every sequence (x_n) *with* $x_n \in I\setminus\{a\}$ *for all* $n \in \mathbb{N}$ *and* $\lim_{n\to\infty} x_n = a$*, we have:*

$$\lim_{n\to\infty} f(x_n) = L$$

Proof. Assume $\lim_{x\to a} f(x) = L$. Let (x_n) be any sequence with $x_n \neq a$, $x_n \to a$.

Given $\varepsilon > 0$, choose $\delta > 0$ from the definition. Since $x_n \to a$, there is some N such that $n \geq N \implies |x_n - a| < \delta$. Then for $n \geq N$, $|f(x_n) - L| < \varepsilon$. Thus $\lim_{n\to\infty} f(x_n) = L$.

For the converse, we prove the contrapositive. If $\lim_{x\to a} f(x) \neq L$, then $\exists\varepsilon > 0$ such that for every $\delta = 1/n$, we can pick x_n with $|x_n - a| < 1/n$ but $|f(x_n) - L| \geq \varepsilon$.

Then (x_n) is a sequence with $x_n \to a$ but $\lim_{n\to\infty} f(x_n) \neq L$. □

This theorem gives us two ways to work with limits:

1. To prove a limit exists: Show that for every sequence approaching a, the function values converge to L
2. To prove a limit does not exist: Find TWO sequences approaching a that give different limits for $f(x_n)$

Example 25 ($\lim_{x\to 0}\sin(1/x)$ revisited)**.** *Consider sequences:*

$$x_n = \frac{1}{2\pi n} \to 0$$
$$y_n = \frac{1}{2\pi n + \pi/2} \to 0$$

Then:

$$f(x_n) = \sin(2\pi n) = 0 \to 0$$
$$f(y_n) = \sin(2\pi n + \pi/2) = 1 \to 1$$

giving different limits for different sequences both converging to 0*, so the limit does not exist.*

5.6 Continuity

One of the main uses of limits is to make precise the intuitive idea that one has about a function being "continuous".

Definition 40 (Continuity)**.** *A function* $f : I \to \mathbb{R}$ *is* <u>*continuous at* $a \in I$</u> *if:*

$$\lim_{x\to a} f(x) = f(a)$$

In ε-δ language: f is continuous at a if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |x-a| < \delta \implies |f(x) - f(a)| < \varepsilon$$

(Notice: we no longer need $0 < |x-a|$; we include $x = a$.)

Example 26. *All the functions above that had limits at* a *were continuous at* a*. The functions without limits were, of course, not continuous.*

Example 27. *The function* $f : \mathbb{R} \to \mathbb{R}$

$$f(x) = \begin{cases} 0, x \neq 0 \\ 1, x = 0 \end{cases}$$

is discontinuous at $a = 0$*.*

We won't have time to say much else about continuity in this course, but it will be absolutely fundamental in many of your future courses.

Chapter 6

Big sets

6.1 Cardinality

Recall the following definition.

Definition 41. *Let X, Y be two sets. We say X, Y have the* <u>*same cardinality*</u>*, and write*

$$|X| = |Y|,$$

if there exists a bijection $f : X \to Y$.

Definition 42. *A set X is* <u>*finite*</u> *if $|X| = |n|$ for some $n \in \mathbb{N}$.*

In this case we typically just write $|X| = n$.

Definition 43. *A set X is* <u>*countable*</u> *if it is finite or $|X| = |\mathbb{N}|$.*

A set which is not countable is called <u>uncountable</u>.

Exercise 62. *Prove that $|\mathbb{N}| = |\mathbb{Z}|$.*

Theorem 4. *The rationals $\mathbb{Q}$ are countable.*

Proof. It suffices to show $|\mathbb{N}| = |\mathbb{Q}^+|$, since $\mathbb{Q}$ can then be enumerated by interleaving the positive rationals, their negatives, and zero.

We arrange the positive rationals in an infinite grid, placing p/q (in lowest terms) at row p, column q:

$$\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \dots \qquad \frac{2}{1}, \frac{2}{2}, \frac{2}{3}, \dots \qquad \dots$$

We then traverse this grid by following successive diagonals (the elements with $p + q = 2$, then $p + q = 3$, and so on). This procedure visits every positive rational exactly once and produces an explicit bijection $\mathbb{N} \to \mathbb{Q}^+$. $\square$

6.1.1 Inequalities

We would also like to compare the sizes of sets when they are not in bijection. This is more subtle than it might first appear.

Definition 44. *Let X, Y be two sets. We write*

$$|X| \leq |Y|,$$

if there exists an injection $f : X \to Y$. We write

$$|X| < |Y|$$

if $|X| \leq |Y|$ and $|X| \neq |Y|$.

Proposition 24. *The relation $\leq$ on cardinalities is a partial order, i.e. it is reflexive, antisymmetric, and transitive. It follows that $<$ is a strict partial order: it is irreflexive and transitive.*

Proof. Reflexivity. The identity map $\mathrm{id}_X : X \to X$ is an injection, so $|X| \leq |X|$.

Transitivity. If $f : X \to Y$ and $g : Y \to Z$ are injections, then the composition $g \circ f : X \to Z$ is an injection (the composition of injections is an injection), so $|X| \leq |Y|$ and $|Y| \leq |Z|$ implies $|X| \leq |Z|$.

Antisymmetry. This is precisely the content of the Cantor–Bernstein theorem stated below.

The properties of $<$ follow immediately: irreflexivity holds because $|X| \not< |X|$ (since $|X| = |X|$), and transitivity of $<$ follows from transitivity of $\leq$. □

Remark 7. *We should be a little careful here: there is no "set of all sets", so $\leq$ is not technically a relation on a set. Nevertheless, the partial-order properties hold for any collection of cardinalities one chooses to compare.*

Theorem 5 (Cantor–Bernstein theorem)**.** *Let X, Y be two sets. If $|X| \leq |Y|$ and $|Y| \leq |X|$, then $|X| = |Y|$.*

Proof. * The proof is surprisingly tricky given how intuitively obvious the statement seems. We give a sketch. Let $f : X \to Y$ and $g : Y \to X$ be injections. Define a nested sequence of subsets $A_0 \supset B_0 \supset A_1 \supset B_1 \supset \cdots$ of X by

$$A_0 = X, B_0 = g(Y)$$

and

$$A_{n+1} = g(f(A_n)), B_{n+1} = g(f(B_n)).$$

We note that $gf : X \to X$ restricts to a bijection $A_n \setminus B_n \to A_{n+1} \setminus B_{n+1}$ for all n.

We claim that the map

$$h(x) = \begin{cases} f(x) & \text{if } x \in A_n \setminus B_n \text{ for some } n, \\ g^{-1}(x) & \text{otherwise} \end{cases}$$

is a well-defined bijection $X \to Y$.

Injectivity: Suppose $h(x_1) = h(x_2)$ but $x_1 \neq x_2$. Since f is injective, we may assume without loss of generality that $h(x_1) = f(x_1)$ and $h(x_2) = g^{-1}(x_2)$.

Then $x_2 = gf(x_1)$ and $x_1 \in A_n \setminus B_n$ for some n. It follows that $x_2 \in A_{n+1} \setminus B_{n+1}$, which is a contradiction.

Surjectivity: Let $y \in Y$. Consider $x = g(y)$. If x is not contained in $A_n \setminus B_n$ for any n, then $y = h(x)$. If $x \in A_n \setminus B_n$ for some n, then $x = gf(x')$ where $x' \in A_{n-1} \setminus B_{n-1}$. Hence $y = h(x')$.

□

Remark 8. *Suppose $f : X \to Y$ is a surjective map. You might expect that then $|X| \geq |Y|$. This is indeed the case, but it will require the introduction of the <u>axiom of choice</u>, as we shall see later in this chapter.*

6.1.2 Cantor's theorem

Theorem 6 (Cantor's theorem)**.** *Let X be a set. Then*

$$|X| < |2^X|.$$

Proof. The map $x \mapsto \{x\}$ is an injection $X \to 2^X$, so $|X| \leq |2^X|$. It remains to show there is no bijection $X \to 2^X$.

Suppose for contradiction that $f : X \to 2^X$ were a bijection. Define

$$T = \{x \in X \mid x \notin f(x)\}.$$

Then $T \in 2^X$, so since f is surjective there exists $y \in X$ with $f(y) = T$. We now derive a contradiction by considering whether $y \in T$ or $y \notin T$.

- If $y \in T$, then by definition of T we have $y \notin f(y) = T$: a contradiction.
- If $y \notin T$, then by definition of T we have $y \in f(y) = T$: again a contradiction.

In either case we reach a contradiction, so no such bijection can exist. □

Theorem 7. *The reals $\mathbb{R}$ are uncountable. In particular, $|\mathbb{N}| < |\mathbb{R}|$.*

i

Proof. By Cantor's theorem, $|\mathbb{N}| < |2^{\mathbb{N}}|$. We will show $|2^{\mathbb{N}}| \leq |\mathbb{R}|$, which gives $|\mathbb{N}| < |\mathbb{R}|$ by transitivity.

We construct an injection $2^{\mathbb{N}} \to \mathbb{R}$. Given a subset $S \subset \mathbb{N}$, encode it as a sequence $(a_n)_{n\geq 0}$ of 0s and 1s by setting $a_n = 1$ if $n \in S$ and $a_n = 0$ otherwise. Map this sequence to the real number

$$\varphi(S) = \sum_{n=0}^{\infty} \frac{a_n}{3^{n+1}} \in [0, 1].$$

Using base 3 (rather than base 2) ensures that distinct sequences give distinct real numbers: if $S \neq S'$ then the first index where the sequences differ contributes a discrepancy of $3^{-(n+1)}$ which cannot be cancelled by the tail, so $\varphi(S) \neq \varphi(S')$. Hence φ is an injection. $\square$

6.2 Continuum hypothesis*

Cantor made the following conjecture, called the Continuum hypothesis: there does not exist a set S such that

$$|\mathbb{N}| < |S| < |\mathbb{R}|.$$

This is a quite natural thing to suppose, and you may have unknowingly been assuming it yourself!

It turns out that the continuum hypothesis is a very strange creature: it is independent of the usual axioms of set theory (ZFC). Gödel showed in 1940 that the continuum hypothesis is consistent with ZFC, and Cohen showed in 1963 (using the technique of forcing) that its negation is also consistent with ZFC. Thus the hypothesis can neither be proved nor disproved from the standard axioms alone. Both of these proofs will have to wait for a more advanced course.

6.3 Axiom of choice and Zorn's Lemma*

Given a set S whose elements are sets, we can form the union

$$\bigcup S = \{x \mid \exists X \in S \text{ s.t. } x \in X\}.$$

Definition 45. *Let S be a set whose elements are non-empty sets. A choice function for S is a function*

$$f : S \to \bigcup S$$

such that $f(X) \in X$ for every $X \in S$.

A choice function makes, for each $X \in S$, a specific choice of an element $x \in X$. Note that if S contained the empty set $\emptyset$, then no choice function could exist, since there is no element to choose from $\emptyset$.

Axiom 4 (Axiom of Choice)**.** *Let S be a set whose elements are non-empty sets. Then S has a choice function.*

This is an axiom, not a theorem: like the continuum hypothesis, its truth or falsehood is independent of the other axioms of set theory.

6.3.1 Consequences of the axiom of choice

Proposition 25. *Let $f : X \to Y$ be a surjective function. Then $|X| \geq |Y|$.*

Proof. We construct an injection $g : Y \to X$. For each $y \in Y$, the preimage $f^{-1}(y) \subset X$ is non-empty since f is surjective. Let $S = \{f^{-1}(y) \mid y \in Y\}$; this is a set of non-empty subsets of X. By the Axiom of Choice, S has a choice function $h : S \to \bigcup S = X$. Define $g(y) = h(f^{-1}(y))$.

To see that g is injective, suppose $g(y) = g(y')$. Then $h(f^{-1}(y)) = h(f^{-1}(y'))$; call this common value x. Since h is a choice function, $x \in f^{-1}(y)$ and $x \in f^{-1}(y')$, i.e. $f(x) = y$ and $f(x) = y'$, so $y = y'$. □

Proposition 26. *A countable union of countable sets is countable.*

Proof. Let $\{A_n\}_{n\in\mathbb{N}}$ be a countable family of countable sets; we wish to show $A = \bigcup_{n\in\mathbb{N}} A_n$ is countable.

Since each A_n is countable, for each n there exists a surjection $f_n : \mathbb{N} \to A_n$ (if A_n is finite of size k, repeat the last element indefinitely). By the Axiom of Choice we may choose such a family $\{f_n\}$ simultaneously. Now define $F : \mathbb{N} \times \mathbb{N} \to A$ by $F(n, m) = f_n(m)$. This is surjective. Since $\mathbb{N} \times \mathbb{N}$ is countable (by an argument analogous to the countability of $\mathbb{Q}$), there is a surjection $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$, and composing gives a surjection $\mathbb{N} \to A$. Hence A is countable. □

Definition 46. *Let X be a partially ordered set. A <u>chain</u> $S \subset X$ is a totally ordered subset, i.e. for every $a, b \in S$ we have $a \leq b$ or $b \leq a$.*

Definition 47. *An element $m \in X$ is called a <u>maximal element</u> if there is no $x \in X$ with $x > m$. (Note: a maximal element need not be greater than all other elements; contrast with a <u>maximum</u>, which satisfies $m \geq x$ for all $x \in X$.)*

Theorem 8 (Zorn's Lemma)**.** *The Axiom of Choice is equivalent to the following statement: if X is a non-empty partially ordered set in which every chain has an upper bound in X, then X has a maximal element.*

6.4 Intervals

Recall we defined earlier the notion of open intervals (a, b) with $a, b \in \mathbb{R} \cup \{\infty\}$. It is also useful to introduce their closed counterparts:

Definition 48. *Let $a, b \in \mathbb{R}$ with $a \leq b$. A* <u>*closed interval*</u> *is a set*

$$[a, b] := \{x \in \mathbb{R} \mid a \leq x \leq b\}.$$

We can also define "half-closed" intervals $[a, b)$, $(a, b]$ and even $[a, \infty)$, $(-\infty, b]$ analogously.

Proposition 27. *The interval $(0, 1)$ has the same cardinality as $\mathbb{R}$.*

Proof. For this case, there are many possible bijections. One elementary way is to consider the function $f : (0, 1) \to \mathbb{R}$

$$f(x) = \frac{2x - 1}{x(x - 1)}.$$

We leave it as an exercise to check that this is a bijection. □

Exercise 63. *Show that all intervals have the same cardinality.*

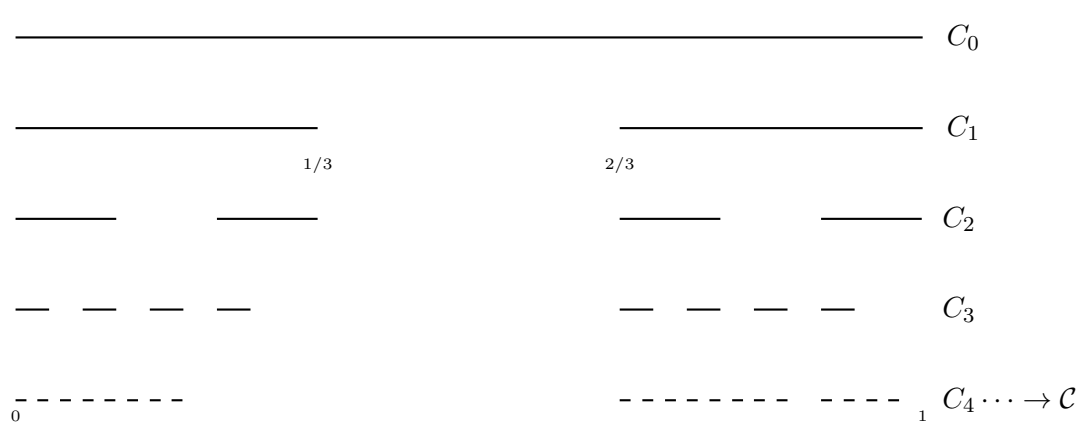
6.5 Cantor set*

Let's try to test the limits of our definitions a bit.

We will now construct a rather interesting set iteratively. Let C_0 be the closed interval $[0, 1]$.

We define a sequence of sets C_n by removing the open middle third of every interval in the previous set:

- <u>Stage 0:</u> $C_0 = [0, 1]$
- <u>Stage 1:</u> Remove $(\frac{1}{3}, \frac{2}{3})$. What remains is $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$.
- <u>Stage 2:</u> Remove the middle thirds of the two intervals in C_1.
  $C_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$.
- <u>Stage n:</u> C_n is the union of 2^n closed intervals, each of length $(1/3)^n$.



The Cantor Set $\mathcal{C}$ is defined as the common intersection:

$$\mathcal{C} = \bigcap_{n=0}^{\infty} C_n$$

Because it is an intersection of nested sets, $\mathcal{C}$ is well-defined. Since each C_n contains the endpoints of its intervals (e.g., $0, 1, 1/3, 2/3, \dots$), these points are never removed and thus $\mathcal{C}$ is clearly non-empty.

An arithmetic perspective (base 3) Recall that every $x \in [0, 1]$ can be written in a ternary (base-3) expansion:

$$x = \sum_{k=1}^{\infty} \frac{a_k}{3^k} = (0.a_1 a_2 a_3 \dots)_3, \quad a_k \in \{0, 1, 2\}$$

Just as $0.999\cdots = 1$ in decimal, we have $0.0222\cdots = 0.1$ in ternary. To ensure uniqueness for the Cantor set, we always choose the non-terminating expansion where possible.

1. Removing the middle third $(1/3, 2/3)$ at Stage 1 removes all numbers whose first ternary digit a_1 must be 1.
2. Removing the middle thirds at Stage 2 removes all numbers whose second ternary digit a_2 must be 1.
3. The Cantor set $\mathcal{C}$ consists exactly of those numbers in $[0, 1]$ that can be written in ternary using only the digits 0 and 2.

6.5.1 The Paradox of Size

Length: almost "empty" set? Let us sum the lengths of the intervals we removed to create $\mathcal{C}$.

- At Stage 1, we remove 1 interval of length $1/3$.
- At Stage 2, we remove 2 intervals of length $1/9$.
- At Stage n, we remove 2^{n-1} intervals of length $(1/3)^n$.

The total length removed is a geometric series:

$$L = \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = \frac{1}{3} \sum_{k=0}^{\infty} \left(\frac{2}{3}\right)^k = \frac{1/3}{1 - 2/3} = 1$$

Since the original interval $[0, 1]$ has length 1 and we removed a total length of 1, our intuition suggests $\mathcal{C}$ should be "empty" or at least very small (like a set of discrete points).

Cardinality: no smaller than $\mathbb{R}$ Despite having "zero length," $\mathcal{C}$ is as large as the entire set of real numbers $\mathbb{R}$.

Theorem 9. *The Cantor set $\mathcal{C}$ has cardinality $|\mathcal{C}| = |\mathbb{R}|$.*

Proof. Let $\mathcal{S}$ be the set of all infinite sequences of 0s and 1s (the power set of $\mathbb{N}$). We know $|\mathcal{S}| = |2^{\mathbb{N}}| = |\mathbb{R}|$.
Define a function $f : \mathcal{S} \to \mathcal{C}$ by:

$$f((s_1, s_2, s_3, \dots)) = \sum_{k=1}^{\infty} \frac{2s_k}{3^k}$$

This maps every binary sequence to a unique ternary expansion consisting only of 0s and 2s. This map is a bijection. Therefore:

$$|\mathcal{C}| = |\{0,1\}^{\mathbb{N}}| = |2^{\mathbb{N}}| = |\mathbb{R}|$$

□

The Cantor set is, in particular uncountable. Despite how small it feels, there are as many points in this "dust" as there are in the entire real line.

6.6 Density*

This section introduces a different notion of "bigness" of a subset of the reals, distinct from cardinality.

Definition 49. *A subset $S \subset \mathbb{R}$ is said to be <u>dense</u> if for all $r \in \mathbb{R}$ and for all $\epsilon > 0$, there exists $s \in S$ such that $|r - s| < \epsilon$.*

That is, a subset of the reals is dense if we can approximate any real number arbitrarily well by elements of that subset.

Proposition 28. *The set of integers $\mathbb{Z} \subset \mathbb{R}$ is not dense.*

Proof. We exhibit a real number that cannot be approximated arbitrarily closely by integers. Take $r = \frac{1}{2}$ and $\epsilon = \frac{1}{4}$. For any $n \in \mathbb{Z}$, we have

$$\left|n - \tfrac{1}{2}\right| = \left|n - \tfrac{1}{2}\right| \geq \tfrac{1}{2},$$

since $n - \frac{1}{2}$ is a half-integer (i.e. of the form $k + \frac{1}{2}$ for $k \in \mathbb{Z}$) and therefore has absolute value at least $\frac{1}{2}$. In particular $|n - \frac{1}{2}| \geq \frac{1}{2} > \frac{1}{4} = \epsilon$ for all $n \in \mathbb{Z}$, so no integer lies within ϵ of r. Hence $\mathbb{Z}$ is not dense in $\mathbb{R}$. □

Proposition 29. *The set of rational numbers $\mathbb{Q} \subset \mathbb{R}$ is dense.*

Proof. Let $r \in \mathbb{R}$ and $\epsilon > 0$. We must find $q \in \mathbb{Q}$ with $|r - q| < \epsilon$.

By the Archimedean property of $\mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > \frac{1}{\epsilon}$, so that $\frac{1}{n} < \epsilon$. Consider the set of integers $\{m \in \mathbb{Z} : m \leq nr\}$; this is non-empty and bounded above by nr, so it has a maximum, which we call m_0. Then by definition $m_0 \leq nr < m_0 + 1$, hence

$$0 \leq r - \frac{m_0}{n} < \frac{1}{n} < \epsilon.$$

Setting $q = \frac{m_0}{n} \in \mathbb{Q}$ we obtain $|r - q| < \epsilon$, as required. □

Exercise 64. *Prove the Archimedean property for $\mathbb{R}$: for any $x \in \mathbb{R}$, $x > 0$, there exists $n \in \mathbb{N}$ with $n > x$.*

We can apply the same definition to any subset of $\mathbb{R}$.

Definition 50. *Let $T \subset \mathbb{R}$. A subset $S \subset T$ is <u>dense in T</u> if for all $t \in T$ and for all $\epsilon > 0$, there exists $s \in S$ such that $|t - s| < \epsilon$.*

6.7 Well-ordering*

Definition 51. *Let $(X, \leq)$ be a totally ordered set. We say $(X, \leq)$ is well-ordered if every non-empty subset $S \subset X$ has a minimum.*

Proposition 30. *Any $n \in \mathbb{N}$ is well-ordered by inclusion.*

Proof. In the von Neumann construction of the natural numbers, each $n \in \mathbb{N}$ is identified with the finite set $n = \{0, 1, \ldots, n-1\}$, ordered by $k \leq m$ iff $k \subset m$ (as sets), which coincides with the usual ordering of natural numbers. We show by induction on n that every non-empty subset of n has a minimum.

<u>Base case</u> $n = 0$: the set $0 = \emptyset$ has no non-empty subsets, so the statement holds vacuously. For $n = 1$: the only non-empty subset of $1 = \{0\}$ is $\{0\}$ itself, which has minimum 0.

<u>Inductive step:</u> Suppose the result holds for n; we prove it for $n+1 = \{0, 1, \ldots, n\}$. Let $S \subset n + 1$ be non-empty. If $S \subset n$, then by the inductive hypothesis S has a minimum. If $n \in S$ and $S = \{n\}$, the minimum is n. Otherwise $S \setminus \{n\}$ is a non-empty subset of n, so by the inductive hypothesis it has a minimum m; since $m \leq n$ for all $m \in n$, this m is also the minimum of S. In all cases S has a minimum, completing the induction. □

Proposition 31. *The natural numbers $(\mathbb{N}, \leq)$ are well-ordered.*

Proof. This follows from (in fact is equivalent to) the principle of induction. Let $S \subset \mathbb{N}$ be non-empty; we show S has a minimum. Suppose for contradiction that S has no minimum. We claim by strong induction that $n \notin S$ for all $n \in \mathbb{N}$.

<u>Base case:</u> $0 \notin S$: if $0 \in S$, then 0 would be the minimum of S (since $0 \leq n$ for all $n \in \mathbb{N}$), contradicting our assumption.

Inductive step: Suppose $k \notin S$ for all $k < n$. If $n \in S$, then since no $k < n$ belongs to S, the element n would be the minimum of S, a contradiction. Hence $n \notin S$.

By induction, $S = \emptyset$, contradicting the assumption that S is non-empty. Therefore S must have a minimum. $\square$

Proposition 32. *The integers $(\mathbb{Z}, \leq)$ are not well-ordered.*

Proof. The subset $\mathbb{Z}$ itself has no minimum element: for any $n \in \mathbb{Z}$ we have $n-1 \in \mathbb{Z}$ with $n-1 < n$, so n cannot be a minimum. Since we have exhibited a subset (namely $\mathbb{Z}$ itself) with no minimum, the ordering is not a well-ordering. $\square$

Proposition 33. *The positive rationals $(\mathbb{Q}^+, \leq)$ are not well-ordered.*

Proof. The subset $\mathbb{Q}^+$ itself has no minimum: for any $q \in \mathbb{Q}^+$, we have $q/2 \in \mathbb{Q}^+$ (since $\mathbb{Q}^+$ is closed under halving) and $q/2 < q$. Hence no element of $\mathbb{Q}^+$ is a minimum. Alternatively, the sequence $1, \frac{1}{2}, \frac{1}{3}, \ldots$ is an infinite strictly decreasing sequence in $\mathbb{Q}^+$, which cannot exist in a well-ordered set. $\square$

Sometimes we don't care about the specific choice of $\leq$; we just want to know that our set X admits some order $\leq$ which is a well-ordering. In this case we say X can be well-ordered.

Proposition 34. *Any countable set can be well-ordered.*

Proof. Let X be countable. Then either X is finite, or it is in bijection with $\mathbb{N}$.

If X is finite, say $X = \{x_1, \ldots, x_n\}$, define the well-ordering by $x_i \leq x_j$ iff $i \leq j$. Every non-empty subset $S \subset X$ then has a minimum, namely the element $x_i \in S$ with smallest index i.

If X is countably infinite, let $f \colon \mathbb{N} \to X$ be a bijection. Define an ordering on X by $x \leq y$ iff $f^{-1}(x) \leq f^{-1}(y)$ in $\mathbb{N}$. This transports the well-ordering of $\mathbb{N}$ to X: given any non-empty $S \subset X$, the set $f^{-1}(S) \subset \mathbb{N}$ is non-empty and hence has a minimum m in $\mathbb{N}$, and $f(m)$ is then the minimum of S in X. $\square$

Theorem 10 (Well-Ordering Theorem)**.** *The Axiom of Choice is equivalent to the following statement: Every set X can be well-ordered.*

Proof. ($\Leftarrow$) Suppose every set can be well-ordered. Let S be a set of non-empty sets; we want to find a choice function for S. Consider the union

$$Y = \bigcup S.$$

Let $(Y, \leq)$ be a well-ordering. For each $X \in S$, let $f(X)$ be the minimal element of X in $(Y, \leq)$; this exists because $X \subset Y$ is a non-empty subset of a well-ordered set. Then $f \colon S \to Y$ is a choice function (since $f(X) \in X$ for each $X \in S$).

($\Rightarrow$) Now assume the Axiom of Choice; we want to show every set X can be well-ordered. We sketch the argument. By the Axiom of Choice applied to the family of all non-empty subsets of X, we obtain a choice function $c\colon \mathcal{P}(X) \setminus \{\emptyset\} \to X$ with $c(A) \in A$ for all non-empty $A \subset X$.

The rough idea is to construct an ordering by continuing the sequence

$$x_0 = c(X), x_1 = c(X \setminus \{x_0\}), x_2 = c(X \setminus \{x_0, x_1\}), ...$$

To really make this work, we need the technique of <u>transfinite recursion</u>, which we will not learn in this class. $\square$

Chapter 7

Sets with structure

What kind of things do mathematicians study? Mathematics is concerned with a huge variety of concepts - sets, manifolds, number fields, statistical ensembles, differential equations, groups, schemes, Hilbert spaces, representations, probability measures, tensor fields, stacks, etc, etc. Many, but not all, of these concepts are based on some sort of data which "enhances" a set into something more. What exactly this "extra data" might be depends a lot on the setting.

This data often allows us to combine elements of the set, compare them, or measure them. For example, from the point of view of set theory, there is no real difference between the set $\{5, \text{dog}\}$ and the set $B = \{0, 1\}$: there is a bijection[1] between them. The same goes for $\mathbb{R}$ and $\mathbb{R} \cup \{\text{Michael}\}$. And yet, we clearly have a feeling that there is a way to add the numbers $0, 1$ or real numbers, but that $\text{Michael} + 1$ is not a particularly sensible idea.

The point is that $\mathbb{R}$ and B are sets with some additional structure – some kind of information beyond just a collection of things (i.e. the elements of a set). Looking at them purely from the set-theoretical point of view is a little bit like closing your eyes and being unable to distinguish between a blank sheet of paper and one with something interesting written on it.

Here are some examples of sets with additional structures:

- $\mathbb{N}$ (natural numbers) has addition and multiplication.
- $\mathbb{R}^2$ (the plane) has vector addition.
- Permutations of $\{1, 2, \ldots, n\}$ have composition.

To describe these more precisely, we should define what kind of object "addition" and "multiplication" are. We define:

[1] Of course, there are multiple bijections between them. Part of the problem is that none of them are obviously better than the other, so even if we wanted to think of 5 and dog as merely different symbols for 0 and 1, it depends on something unnatural – what choice we make.

Definition 52. *Let X be a set. A* <u>*binary operation*</u> *on X is a function*

$$* : X \times X \to X.$$

Usually we just write $*(a, b)$ as $a * b$, in analogy with familiar operations like $+$ and $\times$, or swap the notation out for these in the usual settings.

Thus, a binary operation is a way of "combining" two elements of X to get another element of X. We can think of a binary operation as giving a set some extra structure: for example, it is hard to say there is much structure in the set $\{5, \text{dog}\}$. On the other hand, the set $\{0, 1\}$ naturally has interesting structures on it. We have:

Definition 53 (Boolean integers)**.** *Consider the set*

$$B = \{0, 1\}$$

equipped with:

- *The binary operation $+ : B \times B \to B$ given by $(a, b) \mapsto a + b$ defined by*

$$0 + 0 = 0, 0 + 1 = 1, 1 + 0 = 1, 1 + 1 = 0.$$

- *The binary operation $\cdot : B \times B \to B$ given by $(a, b) \mapsto a \cdot b$ defined by*

$$0 \cdot 0 = 0, 0 \cdot 1 = 0, 1 \cdot 0 = 0, 1 \cdot 1 = 1.$$

We call the triple $(B, +, \cdot)$ (that is, a triple of a set with two binary operations) the <u>Boolean integers</u>.

What we have just described is a set B, together with some operations, functions, relations and elements defined using that set. Starting from these ingredients, we can formulate theorems, proofs and interesting examples.

Exercise 65. *Prove the following fact about the Boolean integers:*

$$\forall x \in B, \exists y \in B(x + y = 0).$$

Here are some more examples.

Example 28. *The familiar operations $+, \times$ give binary operations on $\mathbb{N}$.*

Example 29. *The familiar operations $+, \times$ give binary operations on $\mathbb{Z}$.*

Exercise 66. *Come up with more examples of binary operations.*

Example 30. *The Boolean integers B, equipped with the pair of binary operations defined above, has the structure of a* <u>*ring*</u>[2]*.*

[2] We won't be precise about what this means, but the idea is that it has two binary operations which play well with eachother.

Exercise 67. *Look up what a ring is.*

Although these binary operations in each of these examples actually interact with eachother in a nice way (e.g. distributivity), for now we will focus only on one at a time. We are interested, for now, only in sets equipped with a single binary operation.

7.1 Monoids

Often, we do not start with a specific set. Instead, we leave the choice of set to the reader, but we ask that it come equipped with certain operations, functions, relations and elements satisfying certain properties, which we write down in a definition. Then every example of a set will satisfy every property we prove about the general case. One of the simplest possible kinds of such a definition is the following.

Definition 54. *A monoid $(M, *)$ is a set M together with a binary operation*

$$* : M \times M \to M$$

such that:

- *For all $a, b, c \in M$, $(a * b) * c = a * (b * c)$ (associativity).*
- *There exists $e \in M$ such that for all $a \in M$, $a * e = e * a = a$ (existence of identity).*

Example 31. *We have:*

- $(\mathbb{N}, +)$ *is a monoid (identity* 0*).*
- $(\mathbb{N}, \times)$ *is a monoid (identity* 1*).*
- $(\mathbb{Z}, +)$ *is a monoid (identity* 0*).*
- $(\mathbb{Z}, \times)$ *is a monoid (identity* 1*).*
- *For any set X, the set X^X of all functions $X \to X$ under composition is a monoid (identity $= \mathrm{id}_X$).*
- $(B, +)$ *is a monoid*
- $(B, \cdot)$ *is a monoid*

Example 32. *The following sets with binary operations are not monoids:*

- $(\mathbb{Z}_+, +)$ *(no identity)*
- $(\mathbb{Z}, -)$ *(not associative)*

From the definition, we can prove basic consequences.

Proposition 35 (Uniqueness of identity)**.** *A monoid G has exactly one identity element.*

Proof. Suppose e and e' are both identities. Then $e = e * e' = e'$. □

7.2 Groups

On the one hand, monoids are very general. On the other hand, a monoid is a bit of a weak structure: it is so general that it's hard to prove powerful theorems about them. We would like to study another structure which has a similar flavour, but is more constrained – that of a group. The purpose of defining a group, historically, is to capture the idea of symmetry. That is, if an object has some sort of symmetry, like rotation of a cube, that operation (rotation) should not only be associative and have an identity, but be "invertible".

Definition 55 (Group)**.** *A group is a set G equipped with*

- *An element $e \in G$.*
- *A binary operation $(a, b) \to a \cdot b$.*

such that

- *For all $a \in G$, $e \cdot a = a = a \cdot e$.*
- *For all $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.*
- *For all $a \in G$, there exists $b \in G$ such that $a \cdot b = e$.*

Exercise 68. *Find example of a groupd that is not a monoid.*

From the four axioms alone, we can deduce several fundamental properties.

Proposition 36 (Uniqueness of inverses)**.** *For each $a \in G$, the inverse a^{-1} is unique.*

Proof. Suppose b and c satisfy $a * b = e$ and $a * c = e$. Then $b = b * e = b * (a * c) = (b * a) * c = e * c = c$. □

Proposition 37 (Socks-shoes property)**.** $(a * b)^{-1} = b^{-1} * a^{-1}$.

Proof. $(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * e * a^{-1} = a * a^{-1} = e$. By uniqueness, the inverse must be $b^{-1} * a^{-1}$. □

Proposition 38 (Cancellation laws)**.** *If $a * b = a * c$, then $b = c$ (left cancellation). If $b * a = c * a$, then $b = c$ (right cancellation).*

Proof. Multiply on the left by a^{-1}: $a^{-1} * (a * b) = a^{-1} * (a * c) \Rightarrow (a^{-1} * a) * b = (a^{-1} * a) * c \Rightarrow e * b = e * c \Rightarrow b = c$. $\square$

Example 33. *The following sets with binary operation are groups:*

- $(\mathbb{Z}, +)$
- $(\mathbb{Q}, +)$
- $(\mathbb{Q}_+, \cdot)$